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WordTransABSA: Enhancing Aspect-based Sentiment Analysis with masked language modeling for affective token prediction

Weiqiang Jin [a,b], Biao Zhao [a], Yu Zhang [a,c], Jia Huang [a], Hang Yu [b,*]

[a] School of Information and Communications Engineering, Xi'an Jiaotong University, Xi'an, Shaanxi, 710049, China
[b] School of Computer Engineering and Science, Shanghai University, Shangda Road No. 99., Baoshan, Shanghai, 200444, China
[c] College of Physics & Electronic Information, Yan'an University, Yan'an, Shanxi, 716000, China

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ABSTRACT

In recent years, Aspect-based Sentiment Analysis (ABSA) has been a crucial yet challenging task in recognizing textual emotions from text. ABSA has numerous application across various fields, such as social media, commodity review, and movie comment, making it an attractive area of research. Many researchers are working to develop more powerful sentiment analysis models. Currently, most existing ABSA models use the generic pre-trained language models (PLMs) based fine-tuning paradigm, which only utilizes the encoder parameters while discarding the decoder parameters of PLMs. However, this approach fails to leverage the prior knowledge revealed in PLMs effectively. To address these issues, we investigate the potential of the initial pre-training scheme of PLMs to conduct ABSA and thus propose a novel approach in this paper, namely Target Word Transferred ABSA (WordTransABSA). In WordTransABSA, we propose "Word Transferred LM", a novel sequence-level optimization strategy that transferred target words in sentence into pivot tokens to stimulate better PLM semantic understanding capability. Given a sentence with aspect terms as input, WordTransABSA generates contextually appropriate semantics and predicts the affective tokens on the corresponding positions of the aspect terms. The final sentiment polarity of each aspect term is determined through several sentiment identification strategies that we selected. WordTransABSA takes full advantage of the versatile linguistic knowledge of Pre-trained Language Model, resulting in competitive accuracy compared with recent baselines. The WordTransABSA demonstrates its superiority and effectiveness through extensive experiments in both data-sufficient (full-data supervised learning) and data-insufficient (few-shot learning) scenarios. We have made our code publicly available on GitHub: https://github.com/albert-jin/WordTransABSA.

1. Introduction

Aspect-based Sentiment Analysis (ABSA) is a challenging task in Natural Language Processing (NLP) and has gained increasing dominance in artificial intelligence research. It is a fine-grained form of textual sentiment recognition that aims to identify human attitudes and sentiment polarities towards an aspect term, such as commodities and services, in given unstructured opinionated texts, such as social comments or reviews (Nazir, Rao, Wu, & Sun, 2022; Zhou et al., 2019).

Consider the ABSA example in Fig. 1, where a user comment, *"The perfect **hot pot**, but not warm enough **service**"*, contains two aspect terms, "***hot pot***" and "***service***", associated with the opinion words ***perfect***" and "***not warm enough***", respectively. By accurately recognizing these sentiment associations with aspect topics, valuable insights can be gained, enabling operations supervisors to make smarter strategic decisions.

Over the years, there has been a surge in popularity of PLM-based methods for ABSA (Zhang, Li, & Song, 2019; Zhou et al., 2019), which can harness the powerful semantic representations of PLM encoders such as BERT, BART, T5, and GPT (Brown, Mann, Ryder, Subbiah, et al., 2020; Devlin, Chang, Lee, & Toutanova, 2019; Lewis et al., 2020). By building on top of the encoder of pre-trained language models (PLMs), these models use various downstream classifiers to predict sentiment polarity, which has led to state-of-the-art performance. Furthermore, researchers continue to develop improved sentiment classifiers that can better utilize the semantic representations generated by the encoders of PLMs.

Due to the generic nature of PLM-based fine-tuning paradigms, existing ABSA methods that build on top of the PLM's encoder completely discard the decoder part of the PLM. As illustrated in Fig. 2, we take the

* Corresponding author.
E-mail addresses: weiqiangjin@stu.xjtu.edu.cn (W. Jin), biaozhao@xjtu.edu.cn (B. Zhao), yuu.zhangcs@stu.xjtu.edu.cn (Y. Zhang), huang-jia@stu.xjtu.edu.cn (J. Huang), yuhang@shu.edu.cn (H. Yu).

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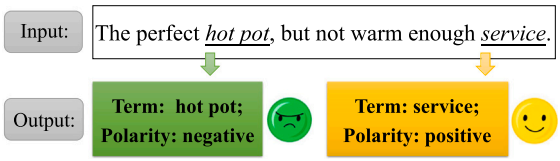


Fig. 1. An example for the aspect-based sentiment analysis task.

BERT as an example for simplicity, which is general and representative. Obviously, the downstream decoding layers (top-right of Fig. 2) adopt a new fine-tunable module that contains external parameters rather than that used in the pre-training "Masked Language Model' task (top-left of Fig. 2), which drops the PLM decoder. Leveraging the partial parameters of the PLMs indicates that the PLM's prior knowledge revealed is not being effectively exploited. Owing to the pre-training and fine-tuning stages belonging to two completely different schemes, the PLM parameter weights distributions have a significant discrepancy. The phenomenon indicates that the optimal fine-tuned ABSA model needs a good parameter optimization, resulting in a time-consuming training procedure and some unstable experimental fluctuations. Moreover, due to the challenge of parameter optimizations, these PLM fine-tuning methods are susceptible to the quality of the training data resources, resulting in the supervised corpora being the common ABSA performance bottleneck. Especially in data-insufficient scenarios such as few-shot learning, sentiment recognition accuracy results decrease significantly.

To this end, we investigate the practical modeling capability and potential of utilizing the Masked LM's pre-training task, "Masked Language Model", to obtain a powerful ABSA model. We introduce a novel model called Target Word Transferred ABSA (WordTransABSA) in this paper. Jin, Zhao, Zhang, Liu and Yu (2023) and Li, Bing, Zhang, and Lam (2019) WordTransABSA takes advantage of the intrinsic characteristic of "Masked Language Model" by transferring the output tokens on the position of target words with several pre-defined sentiment-related pivot tokens. Similar to the "Masked Language Model" task, which predicts the conditional generation probability for all vocabulary tokens of the "[MASK]" position, WordTransABSA aims to predict the sentiment polarity-related affective tokens on the position of an aspect term (also namely target word). WordTransABSA subverts the conventional criterion of PLM-based fine-tuning methods, in which the training process retains the whole sub-modules of Transformer architecture used in PLM pre-training stage. Therefore, WordTransABSA utilizes all the parameters in PLMs, including both the encoder and decoder parts, rather than only using the encoder.

WordTransABSA is an elegant ABSA framework that can be divided into several primary modules: sentiment-related pivot token searching, sentiment-related pivot tokens probability generation, and sentiment words mapping, as depicted in Fig. 4. Firstly, we collect sentiment-related pivot words by searching sentiment lexicons and utilizing Document Retrieval. Secondly, after obtaining sentiment pivot words, we construct a series of training samples. Following the paradigm of "Masked LM" pre-training task, we use the BERT to encode and decode input sequences to generate the targeted sequences we except. Finally, we recover the aspect term sentiment polarities by the pre-defined pivot word mapping strategy.

The comprehensive experiments conducted on WordTransABSA and several comparison baselines showed that WordTransABSA outperformed other standard PLM fine-tuning methods such as BiLSTM-CRF, CasNet, and Exploit-ABSA (Ding, Huang, Jin, Shan, & Yu, 2022; Li et al., 2019; Liu et al., 2018). This suggests that regressing to the PLM's pre-training paradigm is a better solution than the traditional fine-tuning paradigm. Additionally, extensive experiments were performed under different few-shot settings, demonstrating the overwhelming superiority and the superior low-resource convergence capability of WordTransABSA even in data-insufficient scenarios.

Our contributions in this paper can be summarized as follows:

- We propose Target Word Transferred ABSA (WordTransABSA). WordTransABSA subverts the conventional criterion of PLM-based fine-tuning methods by utilizing the entire PLM parameters, to fully exploit the prior knowledge in PLMs.
- Inspired by "Masked LM" pre-training paradigm, we propose "Word Transferred LM", a novel sequence-level optimization strategy that transferred target words in sentence into pivot tokens to stimulate better PLM semantic understanding capability.
- We explore several novel measures, *document retrieval* and *sentiment lexicons search*, to search high-quality sentiment-related pivot tokens to conduct the word substitution optimization.
- The extensive experiments under the data-sufficient scenario (full-data supervised learning) and data-insufficient scenario (few-shot learning) validate the superiority and effectiveness of the WordTransABSA, suggesting that regressing to the PLM pre-training paradigm is a better solution for some specific scenarios like ABSA.
- We have publicly released the implementation of WordTransABSA on Github.[1] We hope that our work can inspire further research in this direction and contribute to the development of ABSA.

The remainder of the paper is organized as follows: Section 2 provides an overview of the prior literature in related fields, with a focus on recent research for ASBA and LM pre-training optimization techniques. Section 4 presents a detailed description of the mechanism of the entire WordTransABSA framework, including a series of formulas. Section 5 provides a comprehensive comparison experiments between WordTransABSA and several ABSA baselines, together with a series of ablation experiments were conducted to demonstrate the technical details and key factors influencing the performance of WordTransABSA. Section 6 delves deeper into the theoretical characteristics of WordTransABSA and other notable approaches. Finally, Section 7 offers a concise summary of the primary contributions of our research and outlines future prospects for further development based on our findings. Code is publicly available at: https://github.com/albert-jin/WordTransABSA.

2. Related works

This work is closely related to existing approaches for Aspect-based Sentiment Analysis (ABSA) (Jin, Zhao, Zhang et al., 2023; Li et al., 2019; Sun, Huang, & Qiu, 2019) and pre-trained language models (Brown et al., 2020; Devlin et al., 2019; Lewis et al., 2020). Therefore, the remaining related works focus on the above two topics.

2.1. PLM-based fine-tuning ABSA methods

The traditional machine learning for ASBA (e.g. SVM, Decision Trees, and Random Forest) (Zhou et al., 2019) has several significant disadvantages that render it impractical as an practical ASBA solution. These approaches suffer from labor-intensive feature definition, dependence on high-quality hand-crafted features, the need for domain expertise and annotations, and limited knowledge transfer to new domains without extensive feature selection efforts.

Recently, many PLM-based ABSA solutions have been proposed with various model augmentation strategies, such as different downstream classifier architectures, pre-trained language model backbones.

The artificial neural network (ANN) (Nazir et al., 2022; Zhou et al., 2019) coupled with the deep representation learning (DRL) scheme, has gained significant attention in mainstream ABSA research. Particularly with the emergence of the well-known BERT (Devlin et al., 2019) and

[1] Code has been released on Github: https://github.com/albert-jin/WordTransABSA.

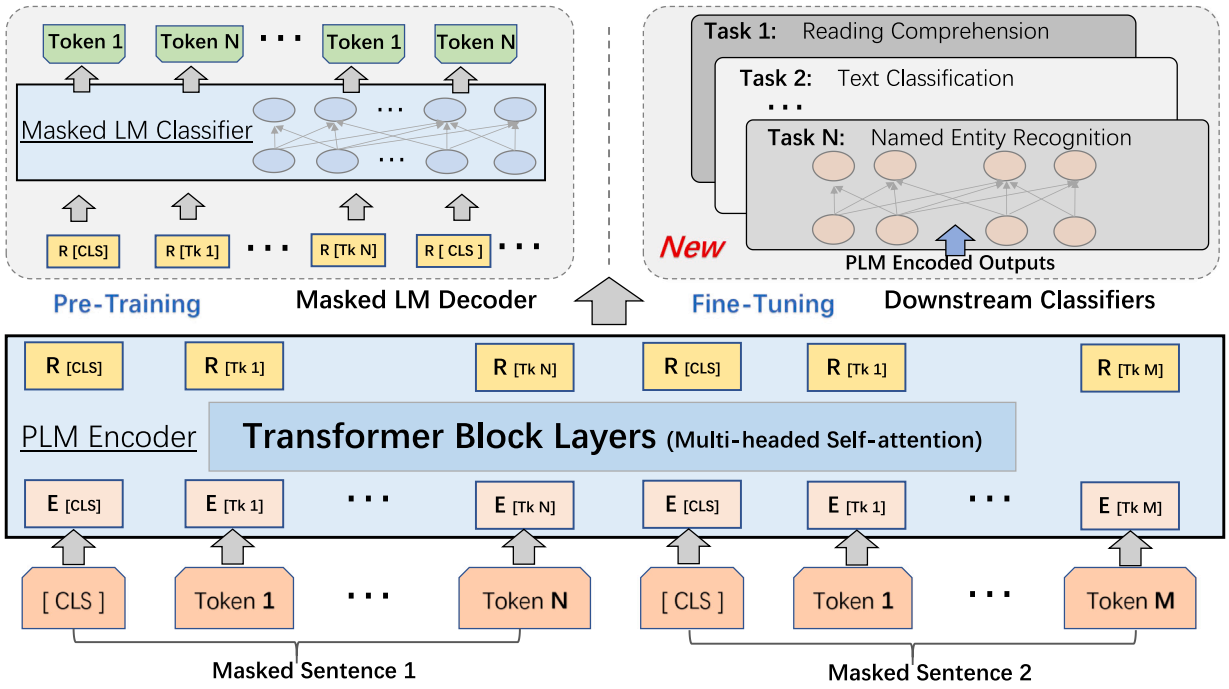


Fig. 2. The concept-level visualizations of the pre-training and fine-tuning stage of most PLMs (e.g. BERT, RoBERTa). The top-left and top-right position shows the pre-training paradigm and the fine-tuning paradigm theoretically, respectively.

GPT-3 (Brown et al., 2020) being proposed, since then, PLM-based fine-tuning ABSA approaches have been continuously achieving state-of-the-art results (Zhou et al., 2019), replacing the pre-trained fixed word embeddings representation methods (e,g. GloVe, Word2Wec) (Mikolov, Chen, Corrado, & Dean, 2013; Pennington, Socher, & Manning, 2014).

Li et al. (2019) conducted a systematic investigation of the practical benefits of PLMs when applied to ABSA, which integrated with various downstream decoding layers, such as linear layer, self-attention networks, and conditional random fields (CRF). Their experiments validate the powerful encoding representation ability of vanilla BERT for ABSA. Xu, Liu, Shu, and Yu (2019) introduce a BERT Post-Training method (BERT-PT) for ABSA inspired by the success of document-level machine reading comprehension (MRC). Zeng, Yang, Xu, Zhou, and Han (2019) developed LCF-BERT, a Local Context Focus mechanism for ABSA, which includes several downstream decoder components such as context features dynamic mask (CDM) and context features dynamic weighted (CDW) layers. Considering the relationship between the two sub-tasks, opinion target word extraction and target sentiment identification, Ding et al. (2022) devise a cascade sentiment analysis model for ABSA (CasNSA) based on their proposed hierarchical tagging decomposition strategy. Jin, Zhao, Zhang et al. (2023) propose DictABSA, which is an external knowledge-infused ABSA framework that injects Oxford dictionary noun-explanatory knowledge to enhance the sentiment prediction ability. Sun et al. (2019) propose CAS-(T)ABSA, which converts ABSA to a sentence-pair classification task by constructing some auxiliary sentences via pre-defined templates. Ling, Yu, and Xia (2022) propose a joint vision-language pre-training framework for the multimodal sentiment recognition task (VLP-MABSA). They utilize position index tags and sentiment one-hot tags to represent the recognition results directly.

In summary, PLM-based ABSA approaches outperform others due to their extensive prior knowledge, making it essential to harness internal PLM knowledge. This paper explores pre-training vs. fine-tuning differences and thus introduces WordTransABSA, a novel ABSA solution mimicking PLM's pre-training paradigm, different from traditional fine-tuning methods.

2.2. Pre-trained language models

Before PLMs, RNNs (e.g., LSTM, GRU, Bi-LSTM) (Zhang et al., 2019) using pre-trained word embeddings (e.g., Word2Vec, GloVe Mikolov et al., 2013; Pennington et al., 2014) were commonly used for processing sequential linguistic data. However, pre-trained word embeddings have limitations in representing polysemous words in different contexts. This led to the development of PLMs, which have achieved outstanding performance across various NLP tasks by learning rich linguistic knowledge from large-scale unsupervised data through self-supervised learning (Han et al., 2021).

Pre-trained language models, like BERT and GPT, are built on the Transformer architecture (Vaswani et al., 2017; Xia, Yu, Wang, Xuan, & Luo, 2023), a non-recurrent seq2seq structure consisting of encoders and decoders. This architecture is composed of stacked blocks, each with multi-head self-attention and position-wise feed-forward layers (Han et al., 2021). BERT, a pivotal pre-trained model introduced in 2018 by Devlin et al. (2019) and Gao, Yu, and Zhang (2022), has millions of parameters and excels at capturing syntactic, lexical, and polysemous disambiguation knowledge from vast linguistic text. It employs a bidirectional deep Transformer and focuses on auto-encoding language modeling during pre-training. In contrast, the Generative Pre-Training (GPT), pioneered by Brown et al. (2020), is a unidirectional generative pre-trained language model. GPT uses auto-regressive language modeling as its pre-training objective, maximizing conditional probabilities based on preceding words. GPTs have hundreds of millions of parameters and, with successive variants like GPT-2 and GPT-3 (Brown et al., 2020), have expanded to over hundreds of billions of parameters, significantly enhancing natural language understanding through self-supervised learning from extensive corpora.

After BERT and GPT, all subsequent PLMs are variants of these two models (Han et al., 2021), which focus on enhancing the model architecture designs and exploring novel pre-training tasks. Zhuang, Wayne, Ya, and Jun (2021) improved BERT by introducing A Robustly Optimized BERT pre-training Approach (RoBERTa). RoBERTa mainly focuses on the listed changes: removing the "NSP" pre-training task; increasing training epochs within larger batch size and more corpora; increasing training sequence input length; and replacing the static

'[MASK]' pattern with a dynamic '[MASK]' pattern. SpanBERT (Joshi et al., 2020) is an enhanced BERT that better represents and predicts spans of text by optimizing several training rules: masking contiguous random spans, and predicting the entire content of masked span through the span boundaries representations. ALBERT is a lightweight BERT (Lan et al., 2020) that achieves better performance while significantly reducing the parameters of the Transformer. Lewis et al. (2020) propose Bidirectional and Auto-Regressive Transformers (BART), a denoising auto-encoder for pre-training sequence-to-sequence models. BART simultaneously generalize the bidirectional encoding scheme of BERT and the left-to-right decoding scheme of GPT.

In the future, we expect more powerful PLMs to emerge alongside evolving fine-tuning techniques. Using PLMs as the foundation for NLP tasks has become a standard practice. BERT is a popular choice in ABSA fine-tuning, but it is crucial to acknowledge that different models can greatly affect performance (Wang, Li, & Liu, 2021; Yuquan, Qinghe, Wenchi, & Wei, 2023).

3. Methodology preliminaries

To better illustrate the model architecture of WordTransABSA and the corresponding conducted experiments, we formulate the related technical concepts in detail, including the task formulation of "*Aspect-based Sentiment Analysis*" and a formulaic expression about the mainstream per-training task of PLMs, "*Masked LM*".

3.1. Aspect-based sentiment analysis

Aspect-based Sentiment Analysis aims to endow its applied model with the capability of accurately recognizing the sentiment for a given aspect term (also named target word) in a given sentence from three sentiment polarities: positive/neutral/negative. It can be briefly formulated as follows: given an input sentence with L-length tokens $S = \{w_i\}_{i=1}^{L} = \{w_1, \ldots, w_i, \ldots, w_L\}$, assuming that this sentence contains K aspect terms $T_K = \{a_j\}_{j=1}^{K}$. Due to its phrase characteristic, each aspect term a_j can be regarded as a text span $a_j = [P_j^{start}; P_j^{end}]$ in the sentence S, where the P_j^{start} and P_j^{end} denote the positions of aspect term's beginning and ending boundaries, respectively. For each assigned aspect term a_j, the Aspect-based Sentiment Analysis task aims to infer a corresponding tag category y_j for a_j from a predefined sentiment polarity set $Y = \{pos, neg, neu\}$ based on the encoded representations of the contextualized semantics in the sentence S.

For example, considering the user comments "*Windows 8* is annoying but *keyboard* responds well to press", this task aims to address the sentiment analysis of target entities, *Windows 8* and *keyboard* and thus identify negative sentiment towards *Windows 8* and positive sentiment towards *keyboard*.

3.2. Masked language model

Masked Language Model is a crucial pre-training task for BERT and other pre-trained language models. Its goal is to endow the adopted LM with massive versatile linguistic knowledge by predicting the original token in the position of the masked token in the input linguistic sequence, based on the embedded context.

Precisely, given a sentence $S = \{w_i\}_{i=1}^{L} = \{w_1, \ldots, w_i, \ldots, w_L\}$ where L denotes the sequence length, the training data generator would mask 15% tokens at random with the unique token "[MASK]" (refer to the original work of BERT Devlin et al., 2019). After the pre-processed sentence $X = \langle SOS\rangle, w_1, \ldots, w_j, \ldots, [MASK], \ldots, w_L, \langle EOS\rangle$ is input into the pre-trained LM, which consists of an **encoder** part and a **decoder** part. As illustrated in Fig. 3, BERT firstly embeds the inputs into initial embeddings E_d^X and then encodes the E_d^X to low-dimension representations using the following formulas (1), (2), and (3).

$$E_d^X = e(X) = [.., e(w_i), \ldots, e([MASK]), \ldots, e(w_j), ..] \tag{1}$$

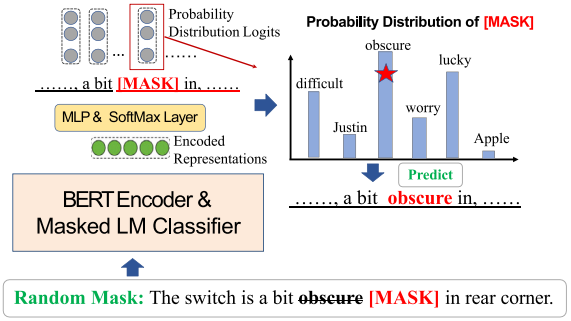


Fig. 3. The concept-level diagram for the pre-trained task of BERT, Masked Language Model.

$$I_d^X = BERT\,Encoder(E_d^X) \tag{2}$$

$$H_d^X = BERT\,Decoder(I_d^X) \tag{3}$$

where e denotes the initial embedding layer, d is the hidden state dimension, $I_d^X \in \mathbb{R}^{X\times d}$ and $H_d^X \in \mathbb{R}^{X\times d}$ denotes the context representations after the BERT encoder and decoder, respectively.

After obtaining the encoded representation H_d^X, a multi-layer perceptron (MLP) and a softmax operation are adopted to calculate the final probability distribution towards the vocabulary of the masked word. This can be calculated as the following formula.

$$\begin{aligned} P(Y \mid X) &= SoftMax\left(MLP(H_d^X)\right) \\ &= SoftMax\left(\mathbf{W}_{ph} \cdot H_d^X + \mathbf{b}_{ph}\right) \end{aligned} \tag{4}$$

where the $P(Y \mid X)$ denotes the conditional probability Y of all input tokens $\{w_i\}_{i=1}^{L}$ when fed with the sentence X, d represents the BERT hidden dimension, which defaults to 768, The MLP denotes the linear transformation (also named prediction head layer), which contains the parameters transform matrix weights $\mathbf{W}_{ph} \in \mathbb{R}^{d\times\|\mathcal{V}\|}$ ($\|\mathcal{V}\|$ represents the vocab size of pre-trained BERT) and the bias $\mathbf{b}_{ph} \in \mathbb{R}^{\|\mathcal{V}\|}$. The output dimension of MLP corresponds to the vocabulary size to display the token probability distributions. The $Softmax$ denotes the softmax operation. Specifically, for the word on "[MASK]" position, assuming that it is located in the kth position, the following formula illustrates the calculating process.

$$P(y_k \mid X) = SoftMax\left(\mathbf{W}_{ph} \cdot h_d^k + \mathbf{b}_{ph}\right) \tag{5}$$

where the $P(y_k \mid X)$ means the token probability distribution of the kth masked word, the y_k denotes the predicted masked word, and h_d^k denotes the encoded representation at the position of the original masked word.

For example, as can be seen from Fig. 3, the sentence "The switch is a bit obscure in rear corner." is randomly masked the word "obscure" with the unique token "[MASK]", and then input into BERT. BERT is pre-trained to choose the most appropriate word "obscure" with the highest conditional probability score for the masked position from the vocabulary.

4. Target word transferred ABSA

In this work, an elegant approach for Aspect-based Sentiment Analysis has been proposed, named Target Word Transferred ABSA (WordTransABSA). As the term suggests, WordTransABSA leverages the automatically substituted words to determine the final sentiment polarities of aspect terms. These substituted words essentially are pre-defined sentiment-related pivot tokens that are semantic-rich with respect to the sentiments (Positive, Neutral, and Negative). Therefore, the entire reasoning procedure involves the following research respects:

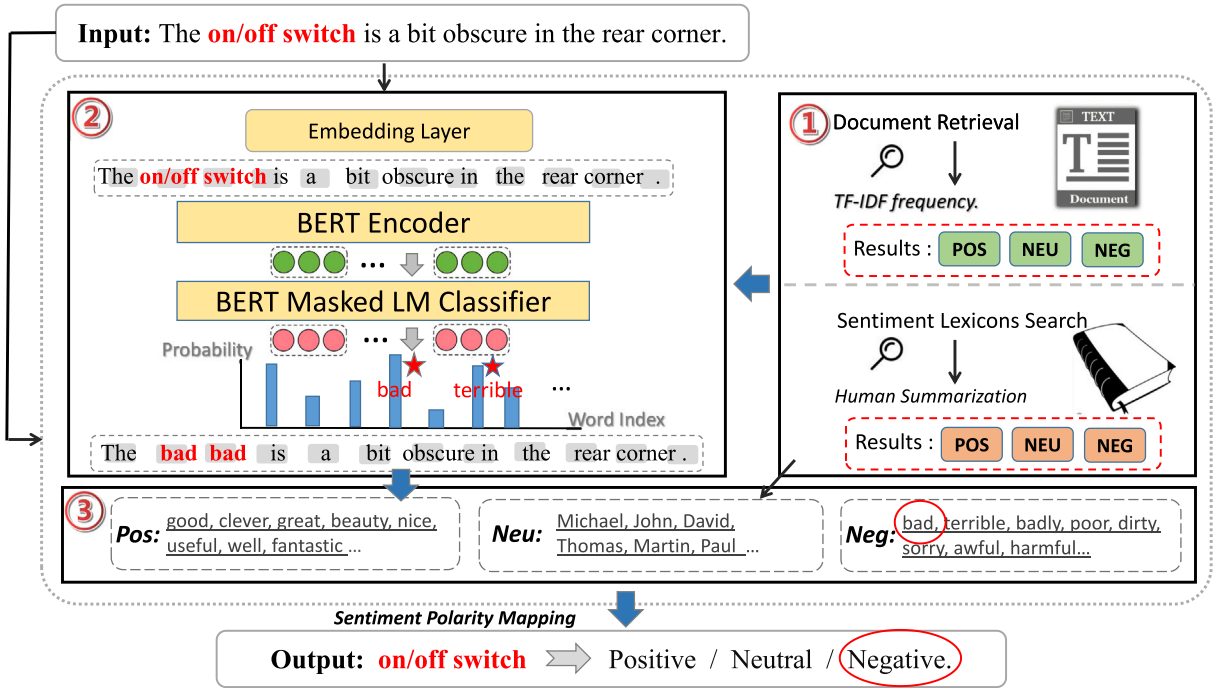


Fig. 4. The model architecture of the proposed WordTransABSA consists of three primary part. Part. (1) denotes the sentiment-related pivot token searching step, including *Document Retrieval* and *Sentiment Lexicons Search*. Part. (2) denotes the probability generation of the sentiment-related pivot tokens based on the BERT's Masked LM; and part. Part. (3) denotes the sentiment words mapping strategy for the final label prediction.

- **Pivot token engineering**: Researching the pivot token selection strategy to discover the sentiment-related words that can better adapt to the adopted LM;
- **WordTrans LM fine tuning**: Researching specific practices for transferring the "Masked LM" pre-training paradigm to a new fine-tuning mode "Word Transferred LM" for better performance on the ABSA task.

Fig. 4 provides a comprehensive visualization of our proposed WordTransABSA framework. In *part.1*, *part.2*, and *part.3* of Fig. 4, we show the procedures for **Pivot token engineering**, **Word Transferred LM fine tuning**, and the selected sentiment-related pivot tokens, respectively. For simplicity, we also provide the concise procedure of pivot token engineering in Algorithm 1 and the concise strategy of WordTrans LM optimization in Algorithm 2.

In conclusion, WordTransABSA firstly substitutes words on the positions of aspect terms in the decoded sequence with sentiment polarity-related affective tokens based on supervised signals. Then, following the original pre-training paradigm "Masked Language Model" of PLMs, WordTransABSA utilizes the paradigm to identify the sentiment polarities of all the aspect terms in the sentence based on the encoded contextual semantic features.

4.1. Sentiment Pivot Token Engineering

In WordTransABSA, the proposed *Word Transferred Language Model* (WordTrans LM) strategy is the key component. To build the core module "WordTrans LM" for WordTransABSA, finding manual-crafted pivot words for word substitution is an essential part that relies on domain expertise knowledge and reasonable selection strategies.

Regarding the aspect of model architecture, our method has certain similarities to Prompt Learning (Liu et al., 2023). There has reached a consensus that manual-crafted templates and manual-selected label words are crucial for prompt-tuning related research. Based on previous work of Gao, Fisch, and Chen (2021), well constituted templates and label words can be incredibly beneficial for the final performance, while unsuitable samples decrease final reasoning accuracy. These Prompt Learning methods are similar to ours, except that the search targets in our work are sentiment-related pivot words instead of templates and label words.

Thus, finding better trigger words that can activate the internal knowledge of the PLMs is significant. The purpose of "Sentiment Pivot Token Engineering" project is to provide high-quality semantic sources for subsequent *Word Transferred Language Model*. Retrieving and selecting pivot words is itself a massive undertaking involving statistics and heavy human involvement. To maintain clarity, we have also provided a concise processing flowchart for "Sentiment Pivot Token Engineering" for the readers, as shown in Algorithm 1.

Inspired by the contextual consistence theory of linguistic corpora (Mikolov et al., 2013; Pennington et al., 2014), we believe that phrases or words containing positive sentiment always co-exist with a semantic-positive context, and vice versa. Therefore, we adopt the following detection strategies to keep a considerable all-around searching scope.

4.1.1. Document retrieval

One practical way to collect sentiment-rich pivot words is by automatically retrieving them from large-scale corpora. The data resources that are of interest are three channels: (1). user comments on social platforms; (2). service evaluations on online shopping platforms; and (3). user expressions from existing ABSA datasets. The detailed processes are described below:

Channel 1 Twitter is the most prevalent social platform in the world, with about 126 million users continuously contributing all kinds of opinions every day. Sufficient user emotional-related textual data can thus be utilized from Twitter. In practice, we download the social messages around the influence of the coronavirus (COVID-19) from the Twitter developer platform,[2] which is a hot topic in recent years.

[2] You can access Twitter developer platform by visiting the following URL: https://developer.twitter.com/en.

Algorithm 1 Sentiment Pivot Token Engineering Steps

Initialize the pre-set candidate words number $\mathbb{N}$ (i.e. 100) for both *Document Retrieval* and *Sentiment Lexicons*;
Initialize the pivot words number o, p, q (i.e. 50) for each sentiment token set, positive/neut-ral/negative;
Input :
Large-scale corpus Twitter $\mathbb{D}_1$ and Amazon $\mathbb{D}_2$, the adopted ASBA datasets $\mathbb{D}_3$, and **NRC** and **OpenHowNet** Sentiment Lexicons $\mathbb{L}$;
Output :
sentiment-related pivot words set for positive/neut-ral/negative: $[\{w_i\}_{i=1}^{o}; \{w_j\}_{j=1}^{p}; \{w_k\}_{k=1}^{q}]$, (o,p,q=50);
for all w' such that $w' \in \mathbb{D}_1 \& \mathbb{D}_2$ & $\mathbb{D}_3$ **do**
    Calculate the $tf-idf$ value $\mathcal{V}_{tf_idf}^{i}$ for each word w_i' based on $\mathcal{V}_{tf_idf}$ for different sentiment topics;
    Ranking words based on sentiment categories;
end for
Manual post – processing :
    filter out non-adjective part-of-speech;
    obviating cross-topics word conflicts;
Return : Top $\mathbb{N}$ ranking candidate pivot words $\mathbb{R}_1$.
Retrieval from **NRC** and **OpenHowNet** $\mathbb{L}$;
    Query the sentiment level of each words;
    Ranking and choose the top-ranking candidate sentiment words $\mathbb{R}_2$;
Affective Words Integration for $\mathbb{R}_1$ & $\mathbb{R}_2$;
Return : Collected sentiment-related pivot word sets: $[\{w_i\}_{i=1}^{o}; \{w_j\}_{j=1}^{p}; \{w_k\}_{k=1}^{q}]$;

Finally, we collected up to three million comments from Twitter, and we have released the retrieved data publicly.[3]

Channel 2 Amazon is the most popular international online shopping platform for people all around the world, which stores numerous user reviews towards various commodities. In practice, we directly use the well-organized corpus, *Amazon Review Data*,[4] which was constructed by Ni, Li, and McAuley (2019). Through their work, *Amazon Review Data* has been processed through high-quality justifications and is thus appropriate to be searched. Specifically, the original corpora include more than 200 million reviews from 1996–2018. Moreover, we only filter the data in the last three years (2016–2018) to keep its timely, with about 60 million reviews.

Channel 3 Moreover, ABSA datasets are natural and effective data-resources for collecting sentiment-rich pivot words due to their massive volume, consisting of the six datasets (Dong et al., 2014; Mitchell, Aguilar, Wilson, & Van Durme, 2013; Pontiki et al., 2014) used in our experiments. In our experiments, we leveraged the training and validation sets of all the adopted datasets to retrieve the target affective words.

Through the channels described above, we have collected abundant intermediate documents for further affective words filtering processes. It should be emphasized that the objective of this procedure is to find comprehensive and highly related affective words that contain rich sentiments. Based on a series of comparisons towards information retrieval, we found that term frequency–inverse document frequency (TF-IDF) can serve as a commonly used and efficient weighting technique for information retrieval and text mining (Grootendorst, 2022; Jin, Zhao and Liu, 2023).

The main idea of TF-IDF is that if a word frequently appears in one article and rarely in others, it is considered to obsess good classification capability and is suitable for text classification. In the ABSA task, the prerequisite of accurate sentiment identification is to have good text classification ability. In our experiments, we adopt the TF-IDF technique to find the words of interest from these documents. The textual corpora collected from **Channel 1** and **Channel 2** can be treated as sentiment-sparse documents, while the original ABSA datasets from **Channel 3** can be treated as sentiment-rich documents. Different sentiment-related topics can be utilized to act as the topic-discovered aspect in TF-IDF. Thus, distinguished from these document topics, we conducted the TF-IDF technique on these documents to retrieve the target affective words.

Given a large-scale corpus $\mathbb{D} = \{S\}_{i=1}^{N}$, in which the $\mathbb{D}$ can be regarded as a set of sentences $\{S\}_{i=1}^{N}$ (sentences count: N), we construct three topics: non-sentiment (sentences mainly from **Channel 1 & 2**, and the ABSA dataset samples which show neutral sentiments), positive-sentiment (sentences mainly from the ABSA datasets which show positive sentiment, and part of comments from **Channel 1 & 2**), and negative-sentiment (sentences mainly from the ABSA datasets which show negative sentiment, and part of comments from **Channel 1 & 2**).

The two channels, **Channel 1 & 2**, mostly are user comments on the Twitter and consumer reviews on the Amazon platform. While it is undeniable that a considerable number of these comments tend to maintain a neutral or relatively subdued sentiment tone, a significant portion of them is infused with vivid sentiment expressions, encompassing both positive and negative sentiments, rather than remaining exclusively neutral.

Based on the above hypothesis, the TF-IDF technique follows the listed procedures:

1. **Term Frequency**: As shown in Eq. (6), we first calculate the term frequency, which is the appearing ratio of the ith word at the ith article. The term frequency TF_{ij} is the appearing number of a word divided by the total number of words in the article. i, j denotes the document-inside and document-outside indexes, respectively; $N_{i,j}$ represents the appearing number of the ith word in the jth document; and $\sum_k N_{k,j}$ represents the total number of words in the jth article.

$$TF_{ij} = \frac{N_{i,j}}{\sum_k N_{k,j}} \tag{6}$$

2. **Inverse Document Frequency**: As shown in Eq. (7), we then calculate the inverse document frequency, which is the universality ratio of a word in all corpora. If the number of documents containing a word in a large-scale corpus is smaller, the IDF of this word would be larger. Furthermore, it demonstrates that the word possesses better classification ability. From this formula, the IDF value of a specific word can be obtained by dividing the total number of documents $\|\mathbb{D}\|$ by the number of documents containing the word $\{j : t_i \in d_j\}$, and then taking the logarithm of the quotient. t_i is the ith word in the vocabulary, and d_j is the jth document in the corpora $\mathbb{D}$. To prevent the denominator from being zero (if the word does not exists in the document), we generally add 1 to the denominator.

$$IDF_i = \log \frac{\|\mathbb{D}\|}{\left\|\{j : t_i \in d_j\} + 1\right\|} \tag{7}$$

3. **TF-IDF**: The term frequency of a word towards a specific document and the inverse term frequency towards the entire corpora can be used to generate the final TF-IDF value through a simple multiplication operation. Finally, each $TF-IDF_{ij}$ represents the importance of a word at the i-index in vocabulary with respect to the entire particular jth document.

$$TF-IDF_{ij} = TF_{ij} \times IDF_i \tag{8}$$

[3] The collected data has been released on Github: https://github.com/albert-jin/WordTransABSA/blob/main/corpora/Twitter-COVID-19.

[4] More details about Amazon Review Data is available at https://nijianmo.github.io/amazon/index.html.

After acquiring the TF-IDF of each word towards the pre-defined topics (positive-sentiment, negative-sentiment, and non-sentiment), we perform data normalization operations and obtain the topic word ranking results for each topic. To filter out target words for different topics, we conduct a series of manual post-processing actions based on affective expertise knowledge. Specifically, we filter out non-adjective part-of-speech and obviating cross-topics word conflicts. Finally, we retain the highly-ranked words, which acts as the sentiment-related pivot words for each pre-defined topic.

The *Document Retrieval* plays a pivotal role in gathering a substantial number of positive and negative pivot words. For instance, within the Twitter corpus, we have identified a spectrum of sentiment-indicative words, including "self-destructive," "weirdest," "banned," "peace," "psycho," "responsible," "smart," "overstate," "obnoxious," and "helpless." The negative term "psycho" was gleaned from comments such as, "This is still the most psycho thing ever," while the negative descriptor "weirdest" was extracted from a sentence like, "The relationship between Donald Trump and Kim Jong-un has to be one of the strangest episodes in US foreign policy."

4.1.2. Sentiment lexicons search

One single word collection lacks universality. It is essential to explore more effective and comprehensive approaches to expand our pivot word sources and enrich the token collections semantically. Another practical way to acquire sentiment-related pivot words is to search sentiment lexicons (i.e. sentiment thesauruses). We refer to two high-quality English sentiment lexicons: the **NRC** Word-Emotion Association Lexicon (Mohammad & Turney, 2013) created by the Institute for Information Technology of National Research Council Canada, and the linguistic ontology knowledge base created by the **OpenHowNet**.[5]

The NRC Emotion Lexicon[6] is manually annotated by crowdsourcing and includes a list of English words and their relationships with two coarse-grained emotions (negative and positive) and eight fine-grained sentiments (surprise, sadness, anger, fear, joy, and anticipation, trust, and disgust). We used the third-party Python package *Emotion Dynamics*[7] to analyze sentiment words towards the NRC Emotion Lexicon. Specifically, we choose a fixed number of top ranking words (e.g. top 100) which are highly associated with the sentiments to act as the filtered targeted words based on the association scores (not associated, weakly, moderately, or strongly associated) of sentiments (positive and negative).

Similarly, the OpenHowNet sentiment lexicon has provided us with well-organized sentiment words that classify English words into positive and negative parts. However, through comprehensive observations, we found that the classification results of this lexicon are not clear enough for direct use due to its inclusion of too many noise words. Thus, we utilized the OpenHowNet sentiment lexicon as a filtering requirement for the affective words output of the NRC Emotion Lexicon. After the sentiment lexicons search procedure, we discovered hundreds of candidate sentiment pivot words for positive and negative sentiments.

4.1.3. Affective words integration

Finally, we integrate these discovered words through weight-based operations. For the positive and negative categories, we select the intersection words in both the document retrieval results and the lexicons search results.

[5] OpenHowNet can be accessed at https://openhownet.thunlp.org/about_hownet.

[6] NRC Emotion Lexicon can be downloaded from: https://www.saifmohammad.com/WebPages/NRC-Emotion-Lexicon.htm.

[7] Emotion Dynamics can be downloaded from: https://github.com/Priya22/EmotionDynamics.

Table 1
The partial examples of the discovered affective words through the sentiment pivot token engineering step.

Polarity	Words examples
Positive	good, excellent, effectual, nice, useful, well, fantastic, ...
Neutral	David, Thomas, Martin, Michael, John, Paul, ...
Negative	bad, terrible, badly, poor, unfortunate, harmful, useless ...

For neutral sentiment-related pivot words, considering the ABSA inherent characteristic (essentially a three-classification problem), the inclusion of neutral pivot words alongside positive and negative ones is essential for a comprehensive sentiment analysis. However, we found that the document retrieval and sentiment lexicons search still have limitations in collecting non-sentiment words in our experiments, which do not provide an ample supply of neutral pivot words. For the neutral pivot words selection, we directly use the words discovered from the document retrieval stage as part of the neutral words, and select the name of a person token as the neutral affective word due to its representation universality. Because names of person are randomly occurring in various contexts and are largely irrelevant to sentiment polarity within the language model which holds reasonable validity.

We retain a fixed number of words for each sentiment polarity to meet the successive experimental requirements. Concretely, we select the top 50 words for the three sentiment polarities in our experiments (150 in total). All the selected affective words have been validated by manual inspection, ensuring the comprehensiveness and quality of the selected sentiment-related pivot tokens. Table 1 gives a brief display of the discovered affective words in the sentiment pivot token engineering. Furthermore, the completed collected pivot words of sentiment categories positive/neutral/negative is displayed in Appendix A.

Collecting sufficient sentiment-related pivot tokens for the three sentiments served two purposes. Firstly, it helped filter better alternative affective words for the Masked LM's sequence-to-sequence learning, enhancing WordTransABSA's encoding-decoding framework. Secondly, these pivot words, rooted in sentiment references, formed a foundation for sentiment classification in subsequent tasks like mapping substitution word probabilities to answers.

4.2. WordTrans LM fine-tuning

After acquiring the sentiment-related pivot tokens, we focus on modeling the affective word transferred language model for fine-tuning (WordTrans LM). As depicted in the left part of Fig. 4, the architecture of our WordTransABSA is unified and uses the BERT encoder as its backbone and the BERT decoder as its sentiment recognizer. From a coarse-grained aspect, BERT embeds and encodes the input sentence into a low-dimension vector representation. The Masked LM decoder of the original BERT is then adopted to predict the substituted affective words (sentiment-related pivot tokens) for the specific aspect terms at the corresponding positions.

The main difference between WordTransABSA and the BERT's Masked LM task is that the original Masked LM task focuses on predicting the original words for the masked words, whereas WordTransABSA slightly modifies the training mechanism, which trains PLM to predict the substituted affective words (sentiment-related pivot tokens) for the corresponding positions of the input aspect terms in a sentence. Throughout the whole model predicting procedure, no extra parameters were added, and no extra loss objectives are reformulate, which preserves the PLM's original pre-training characteristics to the maximum extent.

Concretely, WordTransABSA involves the listed major optimization problems:

1. The strategy for fine-tuning the WordTrans LM using the pre-defined affective words. This highlights how these tokens are integrated into the original Masked LM training mechanism to optimize the WordTrans LM.

Algorithm 2 Word Transferred Language Modeling

Initialize affective word sets for WordTransLM substitution: $[\{w_i\}_{i=1}^{o}; \{w_j\}_{j=1}^{p}; \{w_k\}_{k=1}^{q}]$;
Use: *Average Pooling* & *K-Medoids Clustering*
Return: unique pivot tokens $\{T_{pos}, T_{neu}, T_{neg}\}$ for three sentiment categories, POS/NEU/NEG;
Initialize mapping function $\mathcal{M}: \mathcal{X}$ (source aspect terms) $\rightarrow \mathcal{T}$ (target sentiment pivot tokens);
Given: Training Instances $\mathbb{T} = \{I'\}_{i=1}^{\mathbb{N}}$, $I' = S' \& T'$;
Input :
the original sentence $S = \{x_1, ..., x_k, ..., x_n\}$ with aspect terms $T = \{x\}_j \; j \subset \{1...n\}$;
Output :
the transferred sequence for sentiment analysis $S_{[Trans]} = \{x_1, ..., \mathcal{M}(x_k), ..., x_n\}$, where $\mathcal{M}(x_k)$ is the pivot token (sentiment tag) of x_k;
repeat
  for all $S'_{1,2,3,...} \in \mathbb{T}$ **do**
    Using $\mathcal{M}$ to transfer S' to $S'_{[Trans]}$;
    Compose sequence training pair: $[S'; S'_{[Trans]}]$ for PLM learning using *Masked Language Modeling*;
  end for
until the Word Transferred PLM fully converges;
Inference:
  Fed *WordTransLM* with $S = \{x_1, ..., x_k, ..., x_n\}$;
  Obtain $S_{[Trans]} = \{x_1, ..., \mathcal{M}(x_k), ..., x_n\}$;
  Reasoning of *Affective Words Mapping Strategy*, see Section 4.3;
Return: the final sentiment polarity for each aspect term in S;

2. The strategy for mapping the output of the WordTrans LM to a sentiment polarity. This explains how the final sentiment polarity is predicted based on the string of logits output by the model, using some manually defined criterion for interpretation.

In view of the aforementioned issues, we divide the model architecture of WordTransABSA into two primary modules: *WordTransLM Optimization Objective*, and *Affective Words Mapping Strategy*, which are responsible for solving these issues.

In this section, we will detail the specific working mechanism of *WordTransLM Optimization Objective*. As shown in Fig. 5, this module aims to fine-tune the BERT's internal parameters to generate an ideal decoded sequence when given the original sequence input. Furthermore, to be more intuitive, we also provide the WordTrans LM's optimization procedure in Algorithm 2.

Due to the large volume of filtered affective words, they cannot be utilized directly in WordTransLM. Thus, it is necessary to integrate these affective words as several unique label tokens, which can be used to denote the sentiment polarities.

Formally, prior to the feature propagation of the BERT, we construct the sentiment-related pivot tokens $\{T_{pos}, T_{neu}, T_{neg}\}$ based on the affective word sets $[\{w_i\}_{i=1}^{o}; \{w_j\}_{j=1}^{p}; \{w_k\}_{k=1}^{q}]$, and these generated extra tokens are appended to the BERT vocabulary. In our experiments, we first obtain the initial vector representation $E(w_i)$ of each affective word w_i through the BERT's embedding layer. Then, based on these representations $E(w_i)$, we devise two strategies to obtain the representations of the sentiment-related pivot tokens $E(T)$:

(1) **Average Pooling:** As we know, the average pooling operation is a straightforward and efficient approach for integrating a cluster of multi-dimension vector samples. The averaged vector can always represent a class of vectors that share similar semantic features, which is the sentiment contexts (i.e. pos, neu, and neg) in our experiments.

For example, to obtain the vector representation of the positive sentiment-related pivot token $E(T_j)$, we take the simple vector bit-wise average operation on the positive affective words $\{E(w_i)\}_{i=1}^{M_{pos}}$. The average pooling can be formulated as:

$$E(T_j) = Average(\{E(w_i)\}_{i=1}^{M_j}), \quad j \in \{pos, neu, neg\}. \tag{9}$$

where the $E(\cdot)$ mean the embedding operation by the BERT's embedding layer, and the M_j denotes the affective words number of the jth sentiment polarities (i.e. pos, neu, and neg).

(2) **K-Medoids Clustering:** Since the simple average pooling operation is sensitive to the noise words, the averaged results may suffer from outlier data samples. To generate the ideal representation from the affective words, we believe that a superior clustering algorithm, such as K-Means,[8] and DBSCAN,[9] can be helpful. However, after thorough investigations, we found that these clustering methods are also susceptible to existing noise, which are incorrect affective words. In contrast, the K-Medoids Clustering algorithm can resist noise samples well. Therefore, we choose the K-Medoids[10] to discover the sentiment-related pivot tokens. In addition to the robustness advantage of the algorithm, the K-Medoids Clustering algorithm aims to provide a representative word for each sentiment category's affective words set, instead of giving newly added clustering centers that do not belong to the original affective words.

$$\{E(T_{pos}), E(T_{neu}), E(T_{neg})\} = K - Medoids(\{E(w_i)\}_{i=1}^{N}) \tag{10}$$

where $\{w_i\}_{i=1}^{N}$ denotes all affective words of three sentiment polarities (i.e. $[\{w_i\}_{i=1}^{o}; \{w_j\}_{j=1}^{p}; \{w_k\}_{k=1}^{q}]$, $\{w_i\}_{i=1}^{N}$), the N denotes the sum of the numbers of all affective words (i.e. $o \cup p \cup q$), $\{T_{pos}, T_{neu}, T_{neg}\}$ means the three clustering center points for the three sentiment categories. For details about the corresponding processes, please refer to the online guide book.[11]

After obtaining the vector representations of the sentiment-related pivot tokens, we replace all aspect terms in a given original sentence S with the sentiment-related pivot tokens to obtain the transferred sequence $S_{[Trans]}$. As illustrated in Fig. 5, suppose the pivot token used is "great" here, we replace the aspect term "on/off switch" in the sentence with "great great", where "on/off" and "switch" are replaced with two instances of "great" corresponding to their respective word positions. We apply the above transformations to all original sentences in the ABSA datasets to obtain transferred linguistic sentences. The Word Transferred LM is trained to predict a sentiment pivot token at the position of each word span of an aspect term, which indicates the sentiment polarity of the aspect term. For example, as shown in Fig. 5, the Word Transferred LM predicts two "great" at the positions of "on/off" and "switch".

Formally, suppose there is a mapping function $\mathcal{M}: \mathcal{X}$ (original words of aspect terms) $\rightarrow \mathcal{T}$ (sentiment pivot token of aspect terms). The Word Transferred LM is trained to predict the transferred sequence $S_{[Trans]} = \{x_1, \ldots, \mathcal{M}(x_k), \ldots, x_n\}$ out based on the input sentence (i.e. the original sentence $S = \{x_1, \ldots, x_k, \ldots, x_n\}$). Concretely, all the feature forward processes of the Word Transferred LM are the same as the Masked LM task described in Section 3.2, including the BERT encoder, BERT decoder, and probability distribution prediction. We regard the parameters of the BERT encoder and BERT Masked LM decoder as θ_1 and θ_2, respectively. On top of the downstream classifier, the cross-entropy loss function between the output sequence and the gold-transferred sequence is adopted as the final optimization strategy. Thus, the objective loss function of the Word Transferred LM can be formulated as follows:

$$\mathcal{L}_{WordTransLM} = -\sum_{i=1}^{n} \log P\left(o_i = x_i^{Trans} \mid S\right) \tag{11}$$

[8] More details of K-means: https://en.wikipedia.org/wiki/K-means_clustering.
[9] More details of DBSCAN: https://en.wikipedia.org/wiki/DBSCAN.
[10] More details of K-medoids: https://en.wikipedia.org/wiki/K-medoids.
[11] More details about k-medoids clustering procedure can be accessed at the URL: https://www.javatpoint.com/k-medoids-clustering-theoretical-explanation.

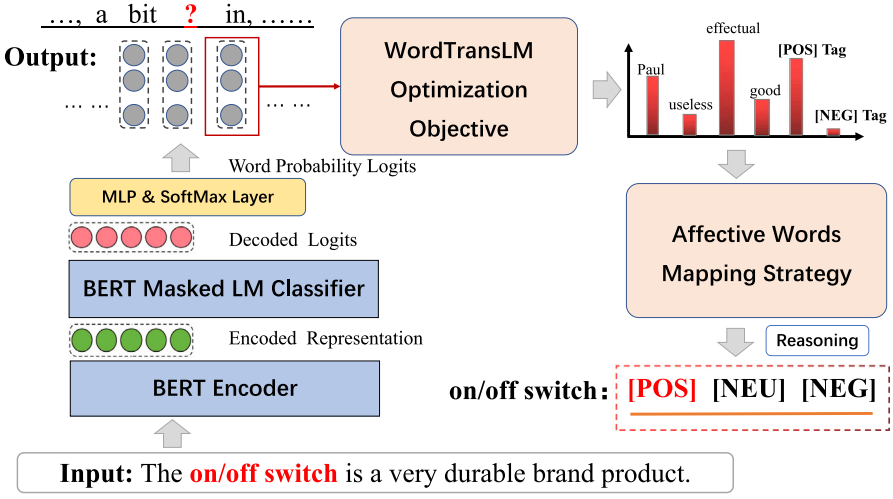


Fig. 5. The detailed reasoning processes illustration of the Word Transferred fine-tuning Language Model.

where o_i denotes the i-index word of the output sequence, the x_i^{Trans} denotes the i-index word of the transferred sequence $S_{[Trans]}$, and S denotes the original input sentence. Note that $P\left(o_i = x_i^{Trans} \mid S\right) = SoftMax(W_{ph} \cdot H_d^S + b_{ph})$, where the H_d^S represents the encoded internal representation of S.

Finally, the model parameters of WordTrans LM are optimized to maximize the probability $P\left(o_i = x_i^{Trans} \mid S\right)$ through the backpropagation algorithm. The above is the modeling procedure of the affective word transferred fine-tuning Language Model.

As illustrated in Algorithm 2, we analyze the time complexity of our Word Transferred Language Model as follows. The previously mentioned pivot token filtering process requires only one computation, and the sequence encoding-decoding process typically follows a sequence-level end-to-end approach, which also requires only one computation. Therefore, the overall time complexity of Algorithm 2 is O(1).

4.3. Affective words mapping strategy

Another crucial problem that should be emphasized is the mapping strategy from the outputs of BERT to the final fine-grained sentiment polarity. As we know, the logits value output by the BERT decoder, which denotes the word probability distributions in the vocabulary, cannot be directly used to determine the final sentiment polarity. Thus, an effective judgment strategy should be devised to conduct further sentiment recognition.

Specifically, given a sentence with a specified aspect term, the Word Transferred LM generates a two-dimensional vector $H_{\|\mathcal{V}\|}^{\mathcal{N}}$, which denotes the predicted probability distributions of the sentence length, where the $\|\mathcal{V}\|$ represents the vocabulary size of pre-trained BERT, and the $\mathcal{N}$ represents the sentence length. Then, we devise two novel strategies based on the observations of the training mechanism of the Word Transferred LM:

1. **Horizontal Detection:** Firstly, we select the predicted word with the highest probability (i.e. $H_{\|\mathcal{V}\|}^{\mathcal{N}}[i-1][0]$) at the starting position of the aspect term (i.e. ith) from the vocabulary. If the predicted word belongs to sentiment-related pivot tokens, the sentiment polarity of this aspect term can be determined;
2. **Vertical Detection:** Different from the *Horizontal Detection* strategy, if the sentiment polarity cannot be determined in the first step, we then focus on the internal sorting outcome (i.e. $H_{\|\mathcal{V}\|}^{\mathcal{N}}[i-1]$) of the start position of the aspect term (i.e. ith) instead of the sentence-level information. Specifically, we select the sentiment-related pivot tokens with the highest ranking probability from the three sentiment-related pivot tokens (pos, neu, and neg) as an indication of the sentiment polarity of the aspect term.

During the experiments, it revealed that specific sentiment polarities had a dominant weight in output probabilities, often exceeding 99%. To address this, a pipeline answer decoding strategy was adopted, involving horizontal and vertical detection steps. The horizontal detection focused on efficiently checking if the word with the highest probability belonged to any sentiment pivot word set, aiming for simplicity but rarely succeeding due to model convergence. Vertical detection was introduced as a supplement, examining the probability distribution of entity word positions in the vocabulary. The pivot word with the highest probability from each sentiment pivot word set was selected, and a probability comparison determined the final sentiment polarity. This pipeline approach ensured both speed and robust performance in cases where horizontal detection failed.

Once again, our approach differs from the previous methods in that we did not introduce any additional parameters to our ABSA solution. Therefore, the LM's original structure is preserved. WordTransABSA breaks the traditional fine-tuning method for PLMs, where the training process retains all the sub-modules used in the pre-training stage. Instead, WordTransABSA employs all the parameters of the original LM to fully utilize the versatile linguistic knowledge in the pre-trained language models.

5. Experiments

To demonstrate the significance of the proposed WordTransABSA with traditional PLM-based fine-tuning methods (Liu et al., 2018; Rietzler, Stabinger, Opitz, & Engl, 2020; Song, Wang, Jiang, Liu, & Rao, 2019; Xu et al., 2019; Zeng et al., 2019), this section introduces in detail the overall experimental procedure and the corresponding experimental analysis. Sections 5.1–5.3 detail some experimental preliminaries; Section 5.4 comprehensively analyze the comparison experiments between WordTransABSA and several baseline methods; Section 5.5 describes the few-shot learning experiments we conducted to quantitatively demonstrate the effectiveness of WordTransABSA under data-insufficient scenarios; and Sections 5.6, 5.8, 5.7 further conduct a series of in-depth experimental explorations of the performance influence of WordTransABSA under various experimental settings and optimization strategies.

5.1. Datasets and evaluation metric

5.1.1. Datasets

For experimental evaluation, we use six real-world benchmarks as the basis for our successive experiments. In all datasets, aspect categories, aspect terms, and sentiment polarities of each expression

sentence are concisely annotated, allowing them to be directly used for ABSA. An detailed overview of these benchmarks are reported in the followings:

- **SemEval-2014:** SemEval-2014 (Pontiki et al., 2014) competition provides two benchmarks related to the restaurant domain and the laptop domain for aspect-based sentiment analysis;
- **SemEval-2015:** SemEval-2015 (Pontiki, Galanis, Papageorgiou, Manandhar, & Androutsopoulos, 2015) workshop organizers annotated a dataset continuation of the released SemEval-2014 restaurant domain benchmark;
- **SemEval-2016:** SemEval-2016, a restaurant domain ABSA dataset proposed by Pontiki et al. (2016), is also a continuation of the SemEval competition;
- **Twitter:** Twitter, collected from the Twitter social platform and annotated by Mitchell et al. (2013). Specifically, in our experiments, we use the latest edition rearranged by Li et al. (2019);
- **ACLShortData:** ACLShortData, a twitter social dataset introduced by Dong et al. (2014) in ACL 14, is annotated based on crowdsourcing through Amazon's Mechanical Turk.

These series of the benchmarks of the international workshop on the Semantic Evaluation (SemEval) competitions (Pontiki et al., 2016, 2015, 2014), are the most widely used domain-specific benchmarks in the literature. They contain the user experiences evaluations of the domains of laptop or restaurant services based on the aspects and respective qualities of their commodities. Note that SemEval-2015 and SemEval-2016 are relatively smaller than the SemEval-2014 benchmark. Moreover, SemEval-2016 covers five domains and involves with eight languages for aspect-based sentiment analysis, in which we consider the English dataset of restaurant domain as the adopted corpus in our experiments. ACLShortData contains about 10,000 English and 30,000 Spanish tweets, in which we consider the English part as the adopted corpus in our experiments.

Furthermore, we summarize the statistical information of these benchmarks in Table 2. In this table, **rest14** and **lap14** respectively denotes the subsets of the restaurant and laptop domains from the SemEval-2014 Competition (task 4) (Pontiki et al., 2014); **rest15** and **rest16** denote the benchmarks released from the SemEval-2015 Competition (task 12) (Pontiki et al., 2015) and the SemEval-2016 Competition (task 5) (Pontiki et al., 2016).

5.1.2. Evaluation metric

To ensure comparability with previous methods, this study evaluates the metric using accuracy, which is the more widely adopted performance evaluation metric in the research of aspect-level sentiment analysis (ASBA), as opposed to precision/recall/F1 (Nazir et al., 2022; Zhou et al., 2019). Generally, a well-implemented ABSA approach should result in a better performance gain, which can be measured by a higher accuracy. Accuracy is a standard metric for evaluating general tasks that can be simplified as label classification tasks. It represents the percentage of correctly predicted samples out of all validation instances and is defined as follows:

$$Accuracy = \frac{Count_T}{Count_N} \tag{12}$$

where the number of correctly predicted samples is denoted as $Count_T$, and the total sample number is denoted as $Count_N$. For each evaluated instance, a prediction is considered correct if the predicted sentiment polarity matches the gold sentiment polarity. Otherwise, it is considered incorrect.

5.2. Comparison baselines

We evaluate the effectiveness of WordTransABSA by comparing it with the following several strong state-of-the-art models.

Table 2
The statistics of several ABSA benchmarks adopted in our experiments.

Dataset	Train	Dev	Test	Total
rest14	1796	181	601	2578
lap14	1308	150	411	1869
rest15	750	82	400	1232
rest16	1113	118	420	1651
Twitter	1687	422	233	2342
ACLShortData	5623	625	692	6940

- **BERT-SPC:** BERT-SPC (Song et al., 2019) refers to the Sentence Pair Classification task of the pre-trained language model, BERT. The input sequence for BERT-SPC is constructed as "[CLS]" + global context + "[SEP]" + [asp] + "[SEP]" when adopted for the ABSA task;
- **BERT-BiLSTM-CRF:** In contrast to the LSTM-CRF model, BERT-LSTM-CRF (Liu et al., 2018) is a competitive model that utilizes pre-trained language model, BERT, instead of pre-trained word embeddings, to acquire character-level word representations;
- **BERT-PT:** BERT-PT (Xu et al., 2019) investigates a new post-training technique applied to the pre-existing BERT model to enhance the fine-tuning performance of BERT for RRC tasks;
- **LCF-BERT:** LCF-BERT (Zeng et al., 2019) is a BERT-shared layer-based approach with a local context focus (LCF) mechanism. It employs the Context Features Dynamic Mask and Context Features Dynamic Weighted layers to capture the extended internal relationships of sentence contexts and aspect terms;
- **BERT-ADA:** BERT-ADA (Rietzler et al., 2020) investigates various fine-tuning approaches by integrating text from additional sources and examining cross-domain training methods.

Correspondingly, we use the notations WordTrans-ABSA$_{avg}$ and WordTransABSA$_{clust}$ to refer to WordTransABSA with the Average Pooling strategy and the K-Medoids Clustering strategy, respectively.

5.3. Experimental settings

The hyperparameter settings used in the experiments are crucial for the outcomes and therefore, the optimal hyperparameter settings have been validated through multiple comparative experiments. Based on the above, we choose the best result to represent the model performance for comparison due to the experimental random fluctuations. These optimal settings have been summarized as follows.

The BERT used in the experiments is implemented as the standard HuggingFace BERT architecture, specifically Bert-Base-Uncased,[12] with 12 transformer blocks, 768 hidden dimensions, and 12 multi-head self-attentions, and approximately 110 million parameters. The BERT-Large model, which has over 340 million parameters, was not used to avoid GPU RAM out-of-memory issues. Although BERT-Large may offer better performance, we chose to prioritize memory efficiency. Other model versions, such as BERT-Mini uncased and BERT-Large cased, can be downloaded from this repository.[13] Notably, we used the pre-trained BERT encoder as-is, without modifying any of its internal parameters or architecture. To ensure precise and accurate performance comparisons, strict hyperparameter control is applied to each WordTransABSA variants during training. Presented in Table 3, we list the optimal hyperparameters for training each of WordTransABSA variants. In this table, LR represents the learning rate for baseline comparison, and $LR_{(few-shot)}$ is for few-shot learning settings; $Epoch$ outside parentheses represent the routine training epochs of full-data supervised learning,

[12] The Bert-Base-Uncased can be downloaded from: https://huggingface.co/bert-base-uncased.

[13] The BERT model variants can be downloaded from: https://huggingface.co/transformers/installation.html#offline-mode.

Table 3
Optimal adopted training time hyper-parameter settings for WordTransABSA and its variants.

Model	LR	LR (few-shot)	LR scheduler	Patience	Batch size	Batch size (few-shot)	Epoch	Optimizer	Random seed	MaxLen
WordTransABSA	0.0001	0.00001	Cosine	5	16	2	20(10)	AdamW	985	128
WordTransABSA	0.0001	0.00001	Cosine	5	16	2	20(10)	AdamW	985	128

Table 4
The ABSA experimental performance statistics (% Accuracy) between the WordsTransABSAs and the SOTA baselines.

	Models	rest14	lap14	rest15	rest16	Twitter	AclShortData
		Acc(%)	Acc(%)	Acc(%)	Acc(%)	Acc(%)	Acc(%)
Baselines	BERT-SPC	86.0	79.3	88.7	87.3	76.0	70.3
	BERT-BiLSTM-CRF	87.3	79.9	**90.3**	88.5	76.4	70.9
	BERT-PT*	85.0	78.1	90.0	87.7	–	–
	LCF-BERT*	87.1	**82.5**	90.1	88.9	76.6	72.7
	BERT-ADA*	**87.7**	79.9	**90.3**	88.2	75.8	71.6
Our models	WordTransABSA$_{avg}$	85.7	78.4	89.1	89.0	76.1	72.5
	WordTransABSA$_{clust}$	87.3	79.7	89.7	**89.3**	**77.4**	**73.4**

while $Epoch$ within parentheses indicates the setting for few-shot learning; $MaxLen$ restricts the maximum sequence length of the PLM input. In our experiments, the dropout rate for all model variants is set to 0.0001 (few shot: 0.00001) for the trainable parameters of BERT and the AdamW optimizer with decoupled weight decay regularization is adopted. We set the random seed to a fixed number, 985, to generate deterministic numbers instead of random numbers for reproducibility. The batch size is set to 16, and the maximum sentence length 128 follow the BERT's initial configurations to ensure regularization instability. Lastly, early stopping is adopted based on the Accuracy evaluated every 5 training epochs on the development sets to prevent overfitting.

5.4. Experimental comparisons

This section mainly focuses on demonstrate the superiority of the proposed WordTransABSA through the conducted comparisons experiments with several SOTA methods. Table 4 displays the statistics for performance comparisons with state-of-the-art (SOTA) baselines.

Excluding traditional machine learning techniques (Nazir et al., 2022; Zhou et al., 2019) and fixed word embedding-based methods (Zhang et al., 2019), all comparative models used for the comparisons are pre-trained language model (PLM)-based fine-tuning methods, which possess superior feature representation capability. Furthermore, for many PLM-based ABSA research, it has been demonstrated that even small modifications to hyperparameters, such as learning rate, can result in significant performance fluctuations for the ABSA task. Therefore, the results reported in this figure, as well as successive figures presented, are the best performances based on multiple experiments carried out under the same settings.

In Table 4, in accordance with the officially released code of most baselines, we instantiate their models and reproduce their performances, and subsequently report the corresponding results. It is worth noting that the experimental results of certain baselines, BERT-PT, LCF-BERT and BERT-ADA, which are marked with an asterisk (*), are quoted directly from their original published literature (Rietzler et al., 2020; Zeng et al., 2019) rather than the results from our reproduced experiments. Concretely, the first and second groups in the table present the results of the comparative baselines and our WordTransABSA models, respectively. The best scores among all models are marked in bold while the second best scores are underlined.

As can be observed from the first three statistical columns of Table 4, the performance of WordTransABSA's variants in terms of accuracy is quite closed to that of other pre-trained language model-based fine-tuning methods. For example, WordTransABSA$clust$ has an absolute accuracy that is slightly lower than that of LCF-BERT and BERT-ADA by around 1.0% to 1.5%. This indicates that the performance of WordTransABSA$clust$ is slightly inferior to these methods in terms of classification accuracy. On the other hand, WordTransABSA$_{clust}$ achieves a slightly higher absolute accuracy than BERT-SPC and BERT-PT by approximately 0.2% to 1.2%, suggesting that it performs slightly better than these models. Overall, these observations demonstrate that the performance of WordTransABSA's variants is competitive with that of other pre-trained language model-based fine-tuning methods.

Based on our performance observations on the Twitter and AclShortData datasets, we have further discovered that our WordTransABSA model surprisingly achieves competitive results than other baseline models. In particular, on the Twitter benchmark, our WordTransABSA$clust$ achieves an absolute accuracy gain of approximately 0.8% compared to LCF-BERT, and an average absolute gain of approximately 1.2% over the five state-of-the-art (SOTA) baselines. Moreover, on the AclShortData benchmark, our WordTransABSA$clust$ model achieves the SOTA performance compared to the other five baselines. Additionally, we found that our WordTransABSA$_{avg}$ model is relatively competitive when trained on both the Twitter and AclShortData datasets compared to the baselines.

To elaborate further, the last two columns of Table 4 compare the performance of the two variants of our WordTransABSAs, WordTransABSAavg and WordTransABSA$clust$. The results indicate that WordTransABSA$clust$ outperforms WordTransABSAavg by a considerable margin. This superiority can be attributed to the selection strategy employed by WordTransABSA$clust$, which is the K-Medoids Clustering. Our experiments demonstrate that the K-Medoids Clustering strategy is better suited to selecting sentiment-related pivot tokens that can effectively represent sentiment polarity, compared to the Average Pooling selection strategy.

Overall, based on the empirical comparison results, it can be concluded that the proposed WordTransABSA has several certain advantages and is competitive over other state-of-the-art baselines in the field of Aspect-Based Sentiment Analysis (ABSA). The WordTransABSA employs the Word Transferred Language Model strategy, which allows it to retain all the parameters of the original LM and fully utilize the versatile linguistic knowledge stored in the pre-trained language model. The results suggest that regressing to the LM pre-training paradigm appears to be a better solution than the traditional fine-tuning paradigm in certain application situations. This Word Transferred approach enables the WordTransABSA to capture more accurate and diverse information from the original pre-trained language model, leading to improved performance in ABSA tasks. In conclusion, the superiority and competitiveness of the proposed WordTransABSA highlight the importance of leveraging pre-trained language models in ABSA tasks.

5.5. WordTransABSA in data-insufficient scenarios

Similar to Prompt Learning (Gao et al., 2021; Liu et al., 2023), our proposed Word Transferred model also retains the entire LM architecture to enable full utilization of the pre-trained model's parameters

Table 5
The performance statistics (% Accuracy) of our WordTransABSA variants and the SOTA baselines under the different few-shot learning settings. All results are averaged over 10 runs to maintain the experimental authenticity.

Few-shots settings	Models	rest14	lap14	rest15	rest16	Twitter	AclShortData
		Acc(%)	Acc(%)	Acc(%)	Acc(%)	Acc(%)	Acc(%)
5-shot	BERT-SPC	0.542	0.539	0.596	0.528	0.547	0.498
	BERT-BiLSTM-CRF	0.575	0.551	0.622	0.532	0.562	0.474
	WordTransABSA$_{avg}$	0.603	0.580	**0.683**	0.619	0.628	0.545
	WordTransABSA*clust*	**0.625**	**0.598**	**0.683**	**0.662**	**0.653**	**0.571**
20-shot	BERT-SPC	0.602	0.575	0.617	0.664	0.624	0.668
	BERT-BiLSTM-CRF	0.640	0.581	0.676	0.679	0.605	0.662
	WordTransABSA$_{avg}$	0.631	0.614	**0.747**	0.675	**0.647**	0.681
	WordTransABSA*clust*	**0.647**	**0.633**	0.726	**0.702**	0.638	**0.695**
50-shot	BERT-SPC	0.729	0.606	0.814	0.702	0.722	0.691
	BERT-BiLSTM-CRF	0.723	0.625	0.808	0.695	0.715	0.677
	WordTransABSA$_{avg}$	0.753	0.663	**0.859**	**0.745**	0.718	**0.698**
	WordTransABSA*clust*	**0.745**	**0.672**	0.842	0.736	**0.724**	0.708

and leverages specific vocabulary conditional generative distributions. We were inspired by the success of Prompt Learning in handling data-insufficient scenarios and decided to investigate whether our WordTransABSA model could perform well under few-shot learning settings in ABSA tasks. This approach involves training the model on a limited amount of annotated data to enable it to generalize to new, unseen data more effectively. By evaluating the performance of our model under such conditions, we aim to demonstrate its ability to handle challenging ABSA tasks in real-world scenarios where labeled data may be limited or expensive to obtain.

Due to experimental constraints, we were unable to consider several baselines, including BERT-PT, LCF-BERT, and BERT-ADA (Rietzler et al., 2020; Xu et al., 2019; Zeng et al., 2019). Therefore, we chose to use two representative methods, BERT-BiLSTM-CRF and BERT-SPC (Liu et al., 2018; Song et al., 2019), as few-shot comparison baselines. This decision was a trade-off between the quality of the demonstrations and practical constraints related to experimental capacity. While we acknowledge the importance of evaluating our model against a broader range of baselines, we believe that the selected comparison methods still provide a reasonable assessment of the performance of WordTransABSA under few-shot learning settings.

Table 5 presents a comprehensive comparison of different few-shot settings. The three comparison groups correspond to the 5-shot, 20-shot, and 50-shot settings, respectively. The results in Table 5 clearly demonstrate that our WordTransABSA variants exhibit superior accuracy performance compared to the baselines. Specifically, in the first group of Table 5, our WordTransABSA*clust* and WordTransABSA*avg* achieved the best and second-best results, respectively, outperforming all baseline methods over the six benchmarks by obtaining an absolute accuracy score gain of about 2%–6%. Conversely, the baseline methods exhibited inferior performance. In the second and third groups of Table 5, our WordTransABSA*clust* and WordTransABSA*avg* also demonstrated promising accuracy scores compared to all baseline methods, particularly on most benchmarks, such as rest14, rest16, AclShotData, and Twitter. However, due to the unique characteristics of the adopted datasets, WordTransABSA*clust* exhibited inferior performance compared to WordTransABSA*avg* under certain conditions, particularly in the 20-shot and 50-shot settings. For example, in the 50-shot learning experiments, WordTransABSA*clust* exhibited a significant accuracy drop of 1.7% and 0.9% on rest15 and rest16, respectively, compared to WordTransABSA*avg*. Nonetheless, based on the comprehensive experiments, it has been demonstrated that our proposed WordTransABSAs exhibit consistency and robustness and remain competitive in most ABSA application scenarios.

From the above phenomena, the experimental results provide strong support for the effectiveness and potential of our proposed WordTransABSA variants in addressing the challenges of few-shot ABSA tasks. Overall, we can induce that the proposed WordTransABSAs have broader implications for low-resource natural language processing and related research fields.

Overall, our experimental results demonstrate that WordTransABSA, which fully leverages the strengths of pre-trained language models and employs a novel word transfer language model strategy, outperforms several state-of-the-art baselines in few-shot learning settings, suggesting that WordTransABSA is particularly effective in data-insufficient scenarios. As we know, traditional fine-tuning methods may suffer from overfitting or insufficient training data, limiting their performance. In contrast, WordTransABSA leverages the rich knowledge learned from pre-trained language models, enabling it to achieve superior performance even when trained on limited amounts of data.

In conclusion, the comparison experiments demonstrate the potential of WordTransABSA as a promising approach for improving the performance of ABSA tasks. Moreover, the few-shot experiments suggest that WordTransABSA can be a useful tool for addressing the challenges of data scarcity in ABSA tasks.

Future research can also focus on optimizing the WordTransABSA model for even greater efficiency and accuracy in data-insufficient scenarios. This may involve investigating other clustering strategies, exploring different combinations of pre-trained language models and fine-tuning techniques, or developing new strategies for selecting pivot tokens. Furthermore, the generalization ability of WordTransABSA to other domains and languages can be examined.

5.6. Applicability of different masked PLMs

Our WordTransABSA fully harnesses the pretraining characteristics of Masked Language Modeling, ensuring that our experiments are not lacking in generality and applicability. In other words, our proposed learning framework remains applicable to all Masked LMs pretrained through the Masked Language Modeling pre-training paradigm. Furthermore, given that BERT is a representative model in the field of classical language models, throughout our experiments, we lean towards using BERT as the foundational language model to showcase the effectiveness and significance of the Word Transferred Language Model training mechanism.

In this section, we conducted a series of supplementary experiments to delve deeper into the universality and applicability of the Word Transferred Language Model strategy under various other Masked PLMs. Specifically, we selected several other forms of PLMs— BERT-Large-Uncased, RoBERTa, DEBERTA, and ALBERT (Devlin et al., 2019; He, Liu, Gao, & Chen, 2021; Lan et al., 2020; Zhuang et al., 2021) — as the basis for WordTransABSA. We then compared these models to the baseline WordTransABSA using BERT-base-uncased to validate the universality of the Word Transferred Language Model across different PLMs.

Table 6
Performance comparison (% Acc) of WordTransABSA with different PLM backbones in full-data learning and few-shot learning for sentiment analysis. All results are averaged over 10 runs to maintain the experimental authenticity.

WordTransABSA variants		rest14	lap14	rest15	rest16	Twitter	AclShortData
WordTransABSA$_{avg}$ on BERT-Base-Uncased (basic implementation)	Full-data training	85.7	78.4	89.1	89.0	76.1	72.5
	Few-shot setting (50)	75.3	66.3	85.9	74.5	71.8	69.8
WordTransABSA$_{avg}$ on BERT-Large-Uncased	Full-data training	86.5	**81.0**	89.8	89.4	77.4	73.3
	Few-shot setting (50)	77.8	67.5	86.4	76.2	73.6	70.3
WordTransABSA$_{avg}$ on RoBERTa	Full-data training	**87.4**	80.3	**90.3**	**90.4**	76.7	**75.7**
	Few-shot setting (50)	77.8	**69.3**	86.9	**78.6**	74.7	71.4
WordTransABSA$_{avg}$ on ALBERT	Full-data training	86.9	80.3	89.8	89.6	**77.6**	74.0
	Few-shot setting (50)	**79.4**	68.9	**87.3**	**78.6**	**75.2**	**72.0**
WordTransABSA$_{avg}$ on DeBERTa	Full-data training	86.3	79.8	89.5	89.4	76.4	73.4
	Few-shot setting (50)	76.7	67.9	86.4	76.7	73.9	70.3

BERT-Large-Uncased[14] (Devlin et al., 2019) is similar to the standard version of BERT, using Masked Language Modeling (MLM) and Next Sentence Prediction (NSP) as pretraining paradigms. This model, which we have adopted, is also uncased, meaning it does not differentiate between English and english. However, it differs from the standard BERT in terms of its capacity and neuron count. This model has the following configuration: 24 layers, 1024 hidden dimensions, 16 attention heads, 336 million parameters.

RoBERTa[15] (Zhuang et al., 2021) is an optimized pretraining approach based on BERT. It enhances BERT by adjusting key hyperparameters, excluding the next-sentence pretraining objective, and employing larger mini-batches and learning rates during training. This refinement results in state-of-the-art performance on various NLP benchmarks (Xie et al., 2022), showcasing the significance of hyperparameter choices and design considerations in language model pretraining (Liu, Wang, Li, Xu, & Li, 2022; Xiao & Du, 2023).

ALBERT[16] (Lan et al., 2020) is a language representation model that improves upon BERT by reducing memory consumption and training speed using two key techniques: splitting the embedding matrix and employing repeating layers among groups. These innovations allow ALBERT to scale effectively, outperforming BERT on various benchmarks like GLUE, RACE, and SQuAD while having fewer parameters.

DeBERTa[17] (He et al., 2021) is an enhanced language model built upon BERT and RoBERTa. It introduces two novel techniques: disentangled attention, which uses separate vectors for content and position, and an enhanced mask decoder for pretraining. These innovations significantly improve model efficiency and performance on various NLP tasks, surpassing RoBERTa-Large even when trained on only half the data (Zhao et al., 2022).

All the pre-trained models used in the experiments are sourced from the transformer ecosystem community provided by the Hugging Face.[18] With the exception of WordTransLM, the concrete implementations of PLM backbones were leveraged through specialized Masked Language Modeling classes provided by Hugging Face: `AlbertForMaskedLM`, `DebertaForMaskedLM`, `ElectraForMaskedLM`, and `RobertaForMaskedLM`, all of which are part of the class `transformers.AutoModelForMaskedLM`[19]

In experimental comparisons, we employed WordTransABSA$_{avg}$ as a baseline model. Specifically, we conducted a series of experiments involving the replacement of its PLM backbone. The statistics of ablated experiments are detailed in Table 6, showcasing performance results under both full-data supervised learning and few-shot learning (50-shot setting) scenarios.

From the experiments, it can be observed that in the full-data supervised learning seconarios, WordTransABSA$_{avg}$ equipped with RoBERTa performs exceptionally well on the Rest14, Rest15, and Rest16 datasets, achieving high accuracies of 87.4, 90.3, and 90.4, respectively. This indicates that RoBERTa excels at capturing semantic information in sentiment analysis, particularly on large-scale datasets, showcasing its robust semantic representational capabilities.

Conversely, in the 50-shot setting, WordTransAB-SA$_{avg}$ adopted with ALBERT exhibits outstanding performance, achieving the highest results in most datasets. This suggests that ALBERT possesses remarkable generalization capabilities, even when data is limited. ALBERT's superior performance in few-shot learning can be attributed in part to its parameter efficiency, parameter sharing strategy, the help of self-supervised loss, and self-attention mechanism, which collectively enable better utilization of limited data.

While WordTransABSA$_{avg}$ equipped with BERT-Large-Uncased both perform well in both full-data supervised learning and few-shot learning tasks, they slightly trail behind RoBERTa and ALBERT. This might imply that larger models, such as RoBERTa and ALBERT, hold certain advantages when dealing with more complex sentiment analysis tasks. DeBERTa demonstrates consistent performance in full-data supervised learning tasks but exhibits a slight decrease in performance in few-shot learning tasks. This suggests that DeBERTa may have relatively weaker generalization capabilities when data is sparse.

In conclusion, different pre-trained language models exhibit distinct strengths in full-data supervised learning and few-shot learning scenarios. RoBERTa excels in data-sufficient training scenarios, while ALBERT showcases remarkable generalization abilities in data-insufficient learning. These results provide valuable insights for selecting the most suitable PLM backbone for specific sentiment analysis. The overall results underscore the universal applicability and effectiveness of WordTransABSA across various Masked PLMs, offering a wider range of choices for practical applications.

It is worth mentioning that self-regressive pre-trained language models, exemplified by GPT, BART, and T5, follow a different approach compared to the mask prediction pre-training paradigm. They leverage the next-word context prediction (namely text generation or conditional generation) pre-training scheme. The latest popular learning paradigm, prompt learning (prompt-tuning), seamlessly accommodates the autoregressive generative traits of these self-regressive PLMs. Undoubtedly, the Word Transferred Language Model strategy we propose can be likened to prompt learning and is considered a native pre-training optimization paradigm tailored for Masked Language Models.

5.7. Ablations of different loss optimization

In the majority of pre-trained language model (LM)-based fine-tuning techniques for diverse downstream language tasks, the dominant

[14] BERT-Large-Uncased: http://huggingface.co/bert-large-uncased.
[15] RoBERTa: https://huggingface.co/roberta-base.
[16] ALBERT: https://huggingface.co/albert-base-v2.
[17] DeBERTa: https://github.com/microsoft/DeBERTa.
[18] Hugging Face transformer: https://huggingface.co/docs/transformers/v4.33.0/en/index.
[19] AutoModelForMaskedLM: https://huggingface.co/docs/transformers/v4.33.0/en/model_doc/auto#transformers.AutoModelForMaskedLM.

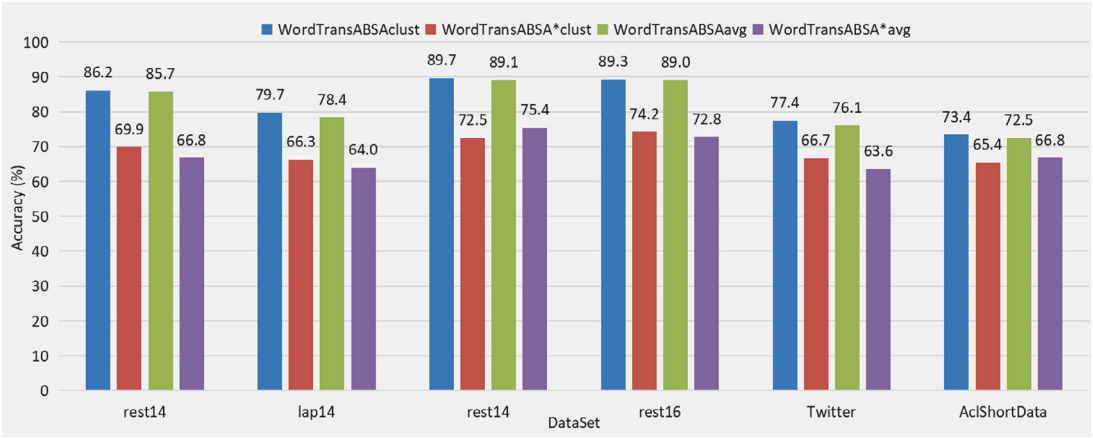


Fig. 6. The accuracy performance statistics, measured in percentage (%), of the ablation experiments conducted on the WordTransABSA variants using the benchmark datasets. WordTransABSA*$clust$ and WordTransABSA*avg denote the variants of WordTransABSA$clust$ and WordTransABSAavg, respectively, which employ the target-level loss optimization strategy.

approach to loss optimization is to solely optimize the loss of the special token, typically the sentence start position tag denoted as "[CLS]" and taken as the optimization target. This approach is referred to as the target-level loss optimization strategy.

However, in the process of decoded optimization in the pre-training task of the Masked Language Model (MLM) of PLMs, all output tokens are considered and all calculated sequence-level losses are integrated for optimization, a strategy known as sequence-level loss optimization. Consequently, this discrepancy between the two optimization strategies cannot be disregarded, as it may have a substantial impact on the final performance of Aspect-Based Sentiment Analysis (ABSA).

Motivated by the notable distinction between the optimizing procedures employed in pre-training and fine-tuning, this section delves deeper into the question of whether the optimization mechanism utilized in the original Masked Language Model task is crucial to the performance of Aspect-Based Sentiment Analysis (ABSA). Specifically, we aim to shed light on the following issues: (1) Can the simultaneous optimization of both aspect terms and contextual words (non-aspect terms) lead to an improvement in ABSA performance? (2) To what extent, quantitatively, does the sequence-level loss optimization strategy enhance the ABSA performance?

To investigate the aforementioned issue, we performed a series of ablation experiments comparing two optimizing strategies as follows:

1. **Sequence-level loss optimization:** This is the original strategy utilized in this work. In this approach, we computed and minimized losses for all tokens in the output sequence, including both aspect and non-aspect terms.
2. **Target-level loss optimization:** In contrast to sequence-level loss optimization, this strategy computed losses solely for the positions of aspect terms in the sequence, while disregarding non-aspect term positions.

In general, combining the aforementioned two optimization strategies, the WordTransABSAs comprise four model variants. This is due to the fact that each of WordTransABSAavg and WordTransABSA$clust$ could produce an additional variant that utilizes the target-level loss optimization strategy outlined above. We refer to these variants as WordTransABSA*$_{avg}$ and WordTransABSA*$clust$, with the asterisk (*) denoting the target-level loss optimization strategy employed. We conducted comprehensive experiments on six benchmark datasets and present the performance results in Fig. 6.

Based on the experimental results shown in Fig. 6, it appears that the performances of WordTransABSAavg and WordTransABSA$clust$ variants, which use the sequence-level loss optimization strategy, are superior to those of their counterparts WordTransABSA*avg and WordTransABSA*$clust$, which utilize the target-level loss optimization strategy. Specifically, the former models show accuracy gains of approximately 10% to 15% over the latter models. In fact, utilizing the sequence-level loss optimization strategy appears to be more effective than the target-level loss optimization strategy for improving ABSA performance.

In summary, our ablated investigation reveals that the optimization mechanism of the original Masked Language Model task is indeed essential to achieving higher performance on the ABSA task. These findings have important implications for researchers and practitioners working on natural language processing tasks, especially those related to sentiment analysis and opinion mining.

5.8. Influence of sentiment pivot tokens selection

In Section 4.1, we have gathered a substantial collection of candidate words. These words are intended to serve as word-level transformation for aspect terms in the target sequence within the Word Transferred LM. The question of how best to use these candidate words to aid the model's optimization process and improve generalization is worthy of careful consideration. Consequently, in this part, we focus on exploring the influence of various sentiment pivot token selection strategies on sentiment identification performance and provide insightful conclusions based on a series of analysis.

To be more specific, our experiments primarily focus on two methods: WordTransABSAavg, which utilizes a bit-wise average strategy to combine vectors from a candidate word set, and WordTransABSA$clust$, which relies on core words obtained through word vector space clustering algorithms. These methods represent two distinct approaches to selecting sentiment substitution words. Here, we introduce three additional word substitution strategies: 1. **WordTransABSA**$vocab_rand$: This strategy involves randomly selecting a word from the vocabulary of the pre-trained language model, which mostly lacks emotion-related semantics relevant to the training process. We label this as WordTransABSA$vocab_rand$; 2. **WordTransABSA**$pivot_rand$: Randomly selecting sentiment words from candidate collections of sentiment pivot words to substitute for the target sequence in the Word Transferred LM. We refer to this strategy as WordTransABSA$pivot_rand$; 3. **WordTransABSA**$avg@10$: We calculate the average of the top 10 sentiment words with stronger emotions. This is denoted as WordTransABSA$avg@10$, which uses fewer sentiment pivot words compared to WordTransABSAavg.

We conducted ten rounds of the evaluation experiments on these ABSA datasets and calculated the average accuracy performance. The experimental results are presented in Fig. 7, which comprises six individual subplots, each corresponding to evaluations on different adopted

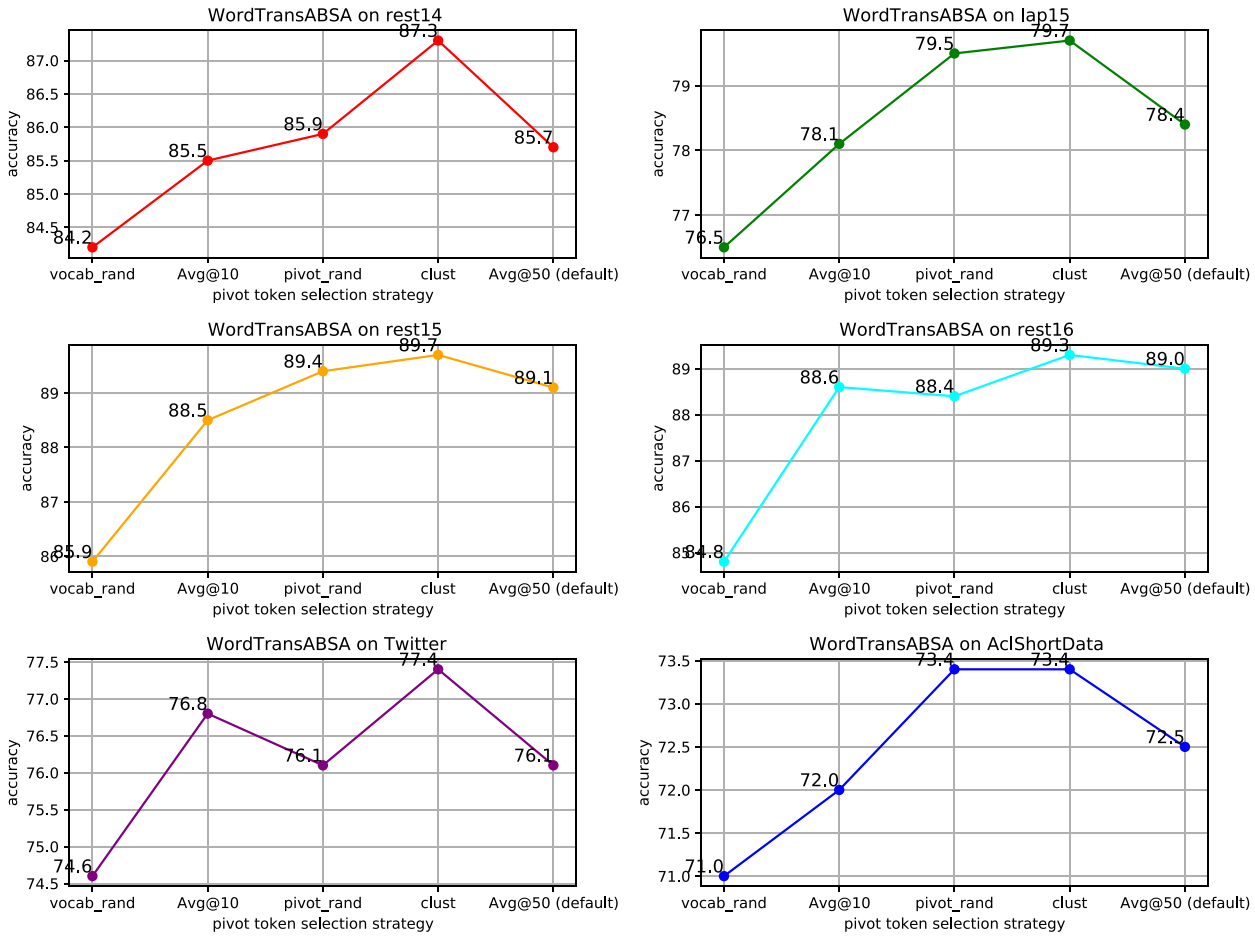


Fig. 7. The Accuracy (%) performance comparisons of different sentiment pivot tokens substitution strategies used to WordTransABSA.

datasets. The X-axis denotes the various pivot token selection strategies, while the Y-axis indicates the accuracy metrics in (% percentage). Notably, WordTransABSA under "clust" and WordTransABSA under "Avg@50" represent the default used word substitution strategies in our primary experiments, specifically, the *K-Medoids Clustering* and the *Average Pooling*, as discussed in Section 4.2.

Firstly, it is evident that selecting sentiment-related elements (whether it is selected sentiment words or operated sentiment vectors) with corresponding sentiment semantics has a significant positive impact on ABSA performance. It can be observed that on the "rest14" and "Twitter" datasets, WordTransABSA$_{vocab_rand}$ performs significantly worse than WordTransABSA$_{pivot_rand}$, with accuracy differences of 1.7% and 1.5%, respectively. Furthermore, WordTransABSA$_{vocab_rand}$ lags far behind other word substitution strategies that incorporate sentiment semantics. Secondly, it can be seen that WordTransABSA$_{avg@10}$ and WordTransABSA$_{avg@50}$ perform similarly across most datasets, with little difference in accuracy. Therefore, we have reason to believe that selecting a small but effective subset of sentiment words is generally sufficient to leverage the advantages of semantic word substitution strategy. Furthermore, WordTransABSA$_{clust}$ outperforms WordTransABSA$_{avg@10}$ and WordTransABSA$_{avg@50}$ in terms of overall performance. This suggests that employing the K-medoids clustering to select the sentiment words is the optimal choice for word substitution strategies in Word Transferred LM.

In summary, in this section, we assess the influence of various sentiment pivot tokens on the sentiment recognition performance of WordTransABSA. Through a comprehensive exploration of diverse word replacement strategies, our experiments have highlighted the essential role played by the choice of an appropriate word replacement strategy as a critical factor in ABSA. Therefore, it is crucial that we devote significant attention to this aspect. The effective utilization of sentiment-related pivot tokens is pivotal for enhancing ABSA performance, and this warrants further consideration in future ABSA-related research.

Furthermore, our study centers on the pivotal role of affective token prediction. Inspired by the perspective of travel demand prediction (Xu, Lv, Li, Sun, & Sheng, 2023) which focuses on urban travel demand prediction, WordTransABSA leads to broader applications. This previous work offers insights into the complexities of various factors, including natural environmental and socioeconomic elements. Importantly, these insights can also enhance affective token prediction, which we have integrated into our sentiment analysis task. We have explored the following key aspects: 1. Correlations with context: Inspired by the literature, we have examined how affective words correlate with contextual elements to refine their selection; 2. Pruning unnecessary affective words: Similar to the literature's use of the RF algorithm to exclude non-significant factors, we have considered excluding irrelevant affective words to improve our model's precision; 3. Mitigating multicollinearity: The concern about multicollinearity in the literature (Xu et al., 2023) prompted us to address similar issues within sentiment analysis. We have explored techniques, including PCA, to address these complexities.

In summary, the above discussion highlights the crucial connection between our affective token prediction of WordTransABSA and the literature (Xu et al., 2023), affirming the potential synergy between the two in advancing the field of sentiment analysis.

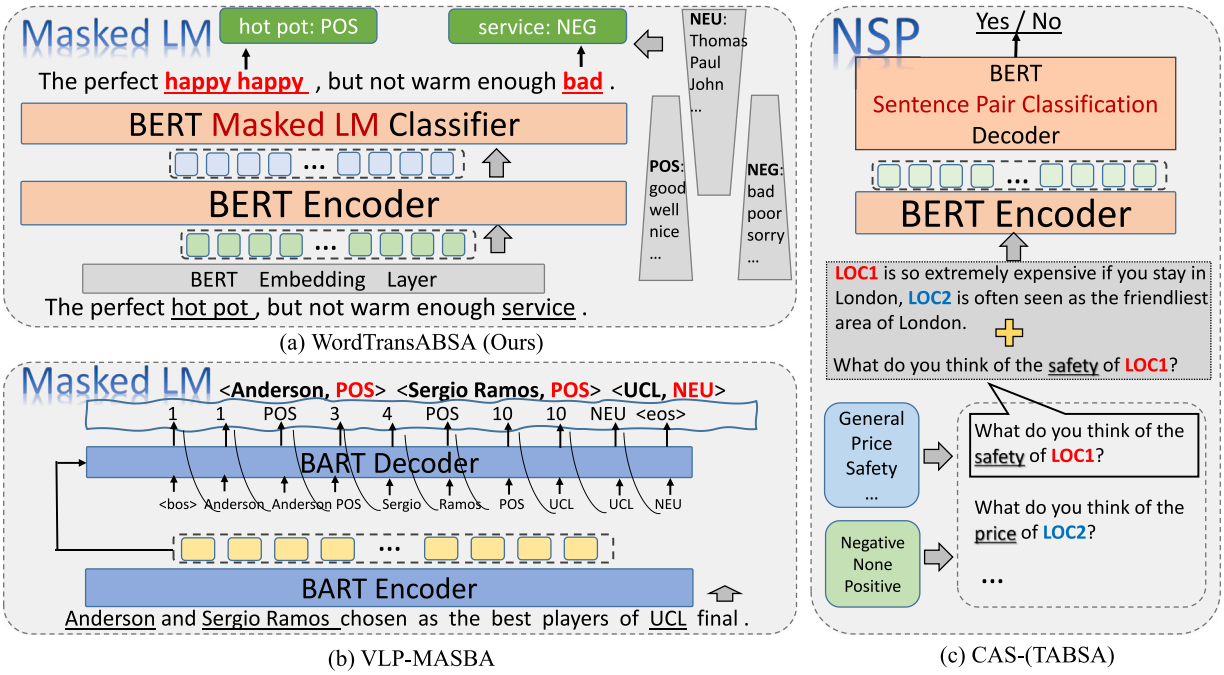


Fig. 8. The model implementation comparisons between our WordTransABSA (a) and the two similar methods, "VLP-MABSA" (b) and "CAS-(T)ABSA" (c). For simplicity, we describe these models by taking several specific sentence examples.

6. Method discussion

In this section, we provide a detailed analysis of the characteristics and advantages of our model from a comprehensive, in-depth theoretical perspective. We do this by the following major points:

- Theoretical analysis, which includes a comparison with existing highly relevant methods;
- Research inspiration and practical implications for successive ABSA research.

6.1. Theoretical comparison analysis

Recently, research similar to our WordTransABSA (Nazir et al., 2022; Zhou et al., 2019) has systematically explored PLM pre-training paradigms' application potential and compared them to original pre-training schemes.

Here, we analyze architecture differences between WordTransABSA and two representative ABSA methods, "VLP-MABSA" and "CAS-(T)ABSA" (Ling et al., 2022; Sun et al., 2019), to get a better theoretical insight into WordTransABSA. "VLP-MABSA" (Ling et al., 2022) uses *Masked Language Model* for predicting target word positions and sentiment polarities in Multimodal ABSA. "CAS-(T)ABSA" (Sun et al., 2019) reformulates TABSA as *Sentence Pair Classification* with an auxiliary sentence. These heuristic approaches return to the original pre-training paradigm, related to WordTransABSA, and differ from prior fine-tuning methods. More details in Fig. 8.

6.1.1. Comparisons with VLP-MABSA

"VLP-MABSA" (Ling et al., 2022) is a vision-language pre-training framework for multimodal ABSA. "VLP-MABSA" focuses on aspect term extraction and aspect-oriented sentiment classification, utilizing Bidirectional and Auto-Regressive Transformers (BART) as its core architecture. As shown in Fig. 8(b), given an original sentence as input, the feature processing follows the BART encoder's general encoding stage. The BART decoder generates the target sequence through the auto-generative procedure. "VLP-MABSA" detects the aspect terms and sentiment polarities by making some post-processing for the decoded sequence by BART. The BART generates a set of sequential tags in a sentence. The targeted tags can be denoted as $Y = [a_1^s, a_1^e, a_1^p, \ldots, a_M^s, a_M^e, a_M^p, \langle eos \rangle]$, where the subscript of a denotes the order of appearance in the sentence of a, and the superscript $\{s, e, p\}$ of the aspect term a determines the start position, the ending position and the sentiment polarity of a. Taking the example sentence "*Anderson and Sergio Ramos chosen as the best players of UCL final.*", the BART expected decoded output could be constructed as $[1, 1, POS, 3, 4, POS, 10, 10, NEU, \langle eos \rangle]$.

The main architecture characteristics of "VLP-MABSA" can be concluded as:

- "VLP-MABSA" directly links BART decoder output to aspect terms and sentiment. This approach requires extensive parameter adjustments, feasible only with ample labeled data, slowing training. In contrast, WordTransABSA maintains Masked LM's original characteristics, yielding faster training and competitive performance in data-scarce scenarios.
- "VLP-MABSA" assumes fixed output order, making sentiment recognition vulnerable to errors. A small mistake in token prediction can disrupt subsequent generation, resulting in fragile and unstable results. For instance, if "VLP-MABSA" misses or adds extra tokens, it may lead to inaccurate interpretation. In contrast, WordTransABSA uses a token substitution strategy that isolates aspect term errors, enhancing model stability in ABSA tasks.

Theoretically, "VLP-MABSA" differs significantly from WordTransABSA: "VLP-MABSA"'s decoding strategy is more intuitive for humans but less interpretable for the neural network. It may not maximize PLM knowledge utilization. In contrast, WordTransABSA aims to predict sentiment-related pivot tokens at aspect term positions, ensuring training consistency for Masked LM. WordTransABSA retains the original expected outputs of other tokens, preserving Masked LM's characteristic. This minimizes costs for faster training and stability.

6.1.2. Comparisons with CAS-(T)ABSA

"CAS-(T)ABSA" (Sun et al., 2019) is a BERT-based fine-tuning method, converting (T)ABSA to the sentence pair classification problem using auxiliary sentences. It provides templates to generate these

auxiliaries automatically, then combined with the original sentence via a special separator, making (T)ABSA a binary classification task. Al predicted sequential labels ($label \in yes, no$) determine the sentiment categories of all aspects. BERT inputs are $Y = [S, [SEP], T_1, T_2, \ldots, A, \ldots, B, \ldots, T_m]$, with $[SEP]$ (special separator), S (original input sentence), T_i (*i-idx* token in a fixed template), A (aspect category), and B (predefined sentiment polarity). In Fig. 8(c), to analyze *price* in "LOC1," an auxiliary like "What do you think of *price* of LOC1" is added, classifying sentiment as positive, neutral, or negative.

The main architecture characteristics of "CAS-(T)ABSA" could be concluded as:

- Different from *VLP-MABSA* and our *WordTransABSA*, "CAS-(T)ABSA" utilizes "Next Sentence Prediction" (NSP) to model the (T)ABSA task as simple binary classification, determining sentence inter-relationships. This approach effectively leverages internal knowledge from LM's pre-training, resulting in superior performance, as shown by experimental results.
- "CAS-(T)ABSA" involves creating many templates for each sentence, which can be time-costing and computationally-consuming. For a sentence with M aspect terms and N aspect categories, "CAS-(T)ABSA" needs to predict all template sentences, resulting in $M \times N$ templates. In contrast, the input complexity of WordTransABSA is O_1, which processes the input sentence once, significantly faster than "CAS-(T)ABSA", which is $O_{M \times N}$ input complexity (processing for each aspect term and each category).

From the above discussion, it is evident that "CAS-(T)ABSA" needs to predict numerous constructed templates, especially in sentences with many aspect terms. In contrast, WordTransABSA, with its Word Transferred Language Models strategy, maps input sentence to output sequences directly, enabling us to determine the sentiment polarity of each aspect term in an end-to-end fashion. Consequently, the inference of WordTransABSA is much quicker, ensuring efficient ABSA solutions in industrial production settings.

Nevertheless, with the continuous advancement of technology, there may be emerging demands for enhanced inference efficiency in the future. To address these potential requirements, we can focus on optimizing and fine-tuning the inference structure of neural reasoning networks.

In recent research, DecomVQANet (Bai, Li, Woźniak, Zhou, & Li, 2021) has emerged as a network compression technique that significantly reduces the reasoning model's parameter volume while maintaining a commendable level of inference performance. Experimental results have shown that this strategy achieves a notable increase in inference speed by reducing parameters through tensor decomposition in deep learning, thereby simplifying computations. DecomVQANet (Bai et al., 2021) exhibits excellent adaptive compression performance with minimal loss in accuracy, placing its performance close to the state-of-the-art levels. Consequently, this decomposition approach presents several advantages over other techniques, making it a promising choice for various applications, including mobile devices, multimedia cabins, public service automation, and more.

In the future, as technology continues to evolve, incorporating techniques like DecomVQANet (Bai et al., 2021) to optimize and enhance our WordTransABSA model's inference efficiency is a promising avenue for future research and development in the field.

6.2. Analysis of optimal input complexity

Here, we analysis the time complexity and input complexity between WordTransABSA, several representative prompt-tuned ABSA methods (such as TemplateBART and SentiPrompt) as well as standard fine-tuned ABSA models (BERT-BiLSTM-CRF, LCF-BERT) to derive conclusions regarding optimal complexity.

1. **TemplateBART:** It employs an exhaustive template-based approach to tackle discriminative tasks. It pre-defines numerous templates, such as "⟨*candidate_span*⟩ *is a* ⟨*entity_type*⟩ *entity*" *for entities and* "⟨*candidate_span*⟩ *is not a named entity*" for non-entities. During template construction, each candidate entity word is allocated to an independent template, resulting in $M + 1$ templates. Candidate word selection involves enumerating from 1-gram to 8-gram for each input sentence, generating 8n spans, which are inserted into the templates. Due to the large template quantity, directly proportional to the entity categories and length of sentence, TemplateBART experiences significantly heavy input complexity. The input complexity can be denoted as $O(M \times N)$, where M is the length of the entire sentence, and N is the number of entity categories. Therefore, when dealing with moderately complex sentences, TemplateBART consumes a substantial amount of training resources and time.
2. **SentiPrompt:** SentiPrompt employs a series of meticulously designed prompting templates, requiring generating templates for each entity word that needs polarity assessment. Consider the sentence: "*A is good, and B is bad.*" SentiPrompt constructs a template like "*[S], Is [X] [Y]? [MASK1], This is [MASK2]*". Here, $[X]$ is dynamically replaced with entities A and B, while $[Y]$ is substituted with corresponding descriptive adjectives. This unique approach enables sentiment polarity evaluation for entities A and B by leveraging the model's predictions for $[MASK1]$ and $[MASK2]$. The template construction process involves exhaustive iteration over every entity and descriptive term within the sentence. Therefore, its time complexity and input complexity scale proportionally with the categories number of A and B, denoted as $O(M \times N)$, where M and N represent the number of target entities and sentiment categories, respectively. Consequently, in cases where the sentence contains a substantial number of entity words, both input and computational complexity experience a notable increase.
3. **Masked PLM based Fine-Tuning:** For standard fine-tuning models using Masked PLMs (such as BERT-BiLSTM-CRF, BERT-SPC, and LCF-BERT), the output dimensions of downstream sentiment polarity classifiers correspond to different polarities, representing various sentiment polarities. A single round of model prediction is insufficient to comprehensively determine the sentiment polarity of all entity words within a sentence. Taking BERT-SPC as an example, this model combines the contextual information of the sentence with the target entity words to create a new integrated context and then uses the PLM to determine the sentiment polarity based on the [CLS] position. However, this approach implies that the input complexity and time complexity of the traditional PLM fine-tuning paradigm are linearly related to the quantity of target entity words in the sentence, denoted as $O(N)$, where N represents the entity words number. Especially, in scenarios complex sentences that along with higher input complexity, these methods are not ideal solutions.

As for our WordTransABSA, the "Word Transferred Language Model" strategy is introduced, which involves using a masked prediction pre-training method to fine-tune the Masked PLM. The learning pattern of this strategy can be likened to the prompt learning paradigm of autoregressive PLMs. At first, the previous pivot token filtering process requires only one computation. And the sequence-level encoding-decoding process of WordTransABSA typically follows a quick end-to-end fashion, which also requires only one computation. Therefore, the overall time complexity and input complexity of WordTransABSA is O(1).

6.3. Research inspiration and practical implications

The innovations of our proposed WordTransABSA are illuminating and have broad applicability. To help readers understand our contributions better, we provide a detailed discussion of the inspiration for our research ideas and the practical implications of our full experiments.

- In this paper, our WordTransABSA's word-level token transformation strategy, based on the Masked LM task, can be seen as a return to the pre-training scheme. This approach can better leverage the PLM itself and achieve superior performance for ABSA. These results demonstrate that, for PLM-based neural networks, retaining the original learning paradigm during the language model's pre-training stage may be optimal and superior to other possible learning paradigms related to various training objectives.
- Prompt-based learning methods need to put in effort to pre-define abundant and appropriate templates to better stimulate the internal knowledge stored inside PLMs. The quality of the templates greatly impacts the final performance of prompt learning, which lacks reasoning stability in building a stable ABSA application.
- The significant superiority of our word transferred strategy is that WordTransABSA can achieve competitive performance without the need for large-scale datasets, which has been fully demonstrated by extensive few-shot experiments. This is an extremely useful approach for scenarios where there is a deficiency of supervised data resource.
- The strategies to optimize the representations on the positions of the substituted aspect terms are especially crucial in the overall modeling processing. According to experimental statistics, a suitable optimization solution for the transferred token representations would effectively improve the ABSA task's performance.

Overall, the idea of combining of the proposed word substitution strategy with the pre-training Masked LM task can be effective for various NLP tasks in different areas (Nazir et al., 2022), such as named entity recognition (Ma et al., 2022), and knowledge-based question answering (Jin et al., 2022).

This approach has universal characteristics that can help solve many NLP application problems. Many related NLP research areas can benefit from the inspiration of our WordTransABSA research presented in this paper.

7. Conclusion and future perspective

Mainstream aspect-based sentiment analysis (ABSA) approaches focus on devising various downstream classification networks, which neglect the potential of the versatile knowledge learned from large unlabeled corpora in PLMs. To address this bottleneck in ABSA research, we propose a novel PLM-based fine-tuning method for ABSA, called Target Word Transferred ABSA (WordTransABSA), which makes full use of the prior knowledge stored in PLMs. WordTransABSA introduces a novel ABSA modeling approach, called the *Word Transferred Language Model* (WordTrans LM) strategy, which is a disruptive training design that is distinct from existing fine-tuning based ABSA solutions. Specifically, WordTransABSA converts the original pre-training "Masked LM" task of the Language Model to the "WordTrans LM" task, which expects the whole utilized PLM to recognize the sentiment polarities of aspect terms by predicting the sentiment-relative pivot tokens from its decoded sequences. WordTransABSA achieves substantial improvement over existing state-of-the-art ABSA methods in the regular training model (full-data setting). Moreover, extensive few-shot comparison experimental results demonstrate the effectiveness and robustness of the proposed WordTransABSA for tackling the data-insufficient scenarios, which particularly possess significance for real-world practical deployment environments.

Due to its versatile applicability, WordTransABSA can be easily adapted to other specific downstream NLP tasks with just some mirror modifications. In the future, we plan to further explore the use of this technique in other tasks, such as named entity recognition (NER) (Shen et al., 2021), document reading comprehension (DRC), and relation extraction (RE). Additionally, building on the success of the recently proposed ChatGPT, we plan to integrate WordTransABSA with ChatGPT to fully exploit the enormous potential of ChatGPT.

CRediT authorship contribution statement

Weiqiang Jin: Conceptualization, Methodology, Software, Visualization, Validation, Project administration, Investigation, Formal analysis, Writing – original draft, Writing – review & editing. **Biao Zhao:** Writing – review & editing, Investigation, Supervision. **Yu Zhang:** Supervision, Funding acquisition. **Jia Huang:** Visualization, Validation. **Hang Yu:** Writing – review & editing, Funding acquisition, Supervision.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: The corresponding author, Prof. Yu, was supported in part by the Shanghai Committee of Science and Technology, China (Grant No.23ZR1423500), and the Major Research Plan of National Natural Science Foundation of China (Project Code: 92167102). This research was also funded by the Shaanxi Key Laboratory of Intelligent Processing for Big Energy Data(IPBED13).

Data availability

Full code to replicate the experiments are available from Github: https://github.com/albert-jin/WordTransABSA. Related used data will be made available on request.

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Appendix A. Sentiment pivot word overview

All sentiment-related pivot words have been showcased in the supplementary materials, Table A.7. However, due to the scarcity of semantically distinct words in emotionally enriched corpora, they are not easy to identify and are not obvious. Identifying sufficient pivot words for neutral sentiment proved to be challenging. Consequently, obtaining a sufficient quantity of neutral pivot words was not achievable. To address this issue, we chose to supplement the neutral category with personal names. This choice stems from the fact that the names of person occur throughout various linguistic contexts and hold minimal relevance to sentiment polarity within the language model, thus ensuring reasonable validity.

Table A.7
An overall view of collected affective words for three sentiment categories, positive/neutral/negative.

Sentiment polarity	Positive	Neutral	Negative
Collected pivot words	good, excellent, effectual, nice, useful, successful, share, enrich, create , valued, helpful, innovative, warm, admired, creative, friendly,loving, tender, sincere, respected, nice, supportive, appreciated, innovate, peaceful, clean, fantastic, love, delighted, wonderful, enough,dream, inspire, beautiful, live, optimistic, hopeful, improve,well, productive, powerful, free, spirit, cheerleader, fair, expect, great,handsomely, widespread, worth, confident,	amid, TV, school, tea, walk,window, paper,sleep, buy, run, explore,study, eat, watch, drink, do, build, David, Thomas, Martin, Michael, John, Paul, Sophia, James, Emma, Charlotte, Joe, Ella, Zoey, Lucy, Hazel, Lily, Sofia, Luna, Nova, Adeline, Susan,Adeline, David, Elizabeth, Linda, Bella, Skylar, Charlotte, Trump, Biden,Mary, Aria, Melody,	bad, terrible, badly, poor, unfortunate, harmful, useless,nervous,self-destructive, weak, cannot,banned, overstate, helpless, weirdest, obnoxious, psycho, regret, nonsense, dissent, crackdown, overruled, shit, difficult, sacrificed, lose, frustrating, disgrace, humbled, subservient, censured, worst, relentless, conflict, poor, unsatisfactory, awful, horrible, unlucky, damaging,worthless, ineffective, worry, angry, pessimistic, fearful, lazy, hate, doubt, boring,

Appendix B. Affective words mapping strategy: Code and runtime insights

To enhance readers' comprehension of the *Affective Words Mapping Strategy* method, we offer a detailed exploration of its code implementation. In this section, we delve into the basic implementation of WordTransABSA, WordTransABSA$_{avg}$, which represents an average pooling strategy-based substituted vector acquisition (have illustrated in Section 4.2, other model variants are similar). As depicted in Fig. B.9, we present a pivotal code excerpt that harnesses the *Affective Words Mapping* strategy.

Firstly, this code segment employs conditional statements (**if-else**) to execute the pipeline decoding operations. The initial several lines are primarily dedicated to *Horizontal Detection*, followed by subsequent several lines focusing on the execution logic of *Vertical Detection*.

It is essential to emphasize that all this decoding logic is based on the direct tensor output of the Python object '*AutoModelForMaskedLM*' (variable: **eval_outputs**), which also corresponds to the output of WordTransLM. To provide a more intuitive understanding, Fig. B.10 presents a runtime snapshot of WordTransLM (variable: **plm_model** in this figure). Within this figure, a partial visualization of the variable: **eval_outputs** is captured. It is worth noting that the '**logits**' attribute within **eval_outputs** contains the target output of WordTransLM. This tensor has dimensions (2, 36, 28 999), representing batch size (Note that this case belongs to few-shot learning, thus we adopted a batch size of 2.), the maximum sequence length for this batch (In contrast to the experimental preset maximum sequence length of 128, it here refers to this batch's actual longest sequences.), and the size of the word vocabulary for PLM, respectively. The interpretation of this variable can be explained as follows: within an independent prediction batch (with a batch size of 2 instances), for each position in the sequences within this batch (sequence length: 36), WordTransLM predicts a probability distribution for each position in the transferred sentence (post-substitution sequence), indicating the likelihood of each word belonging to the PLM word vocabulary. In the upper right of Fig. B.10, you can also find a partial visualization of this direct output of WordTransLM, **eval_outputs**.

Returning to Fig. B.9, the variable **pred_tags** is determined by selecting the indexes with the highest logits along the last dimension of the variable: **eval_output.logits**. Its significance lies in being the maximum probability decoded sequence produced by WordTransLM, as depicted in the operational logic of '.argmax(dim=-1)' in Fig. B.10. Through Horizontal Detection, a comparison is drawn between the starting position of the aspect term and the candidate pivot tokens. In the case of WordTransABSA$_{avg}$, these pivot tokens are three sentiment representation vectors newly incorporated into the PLM word vocabulary, identified by indices 28 996, 28 997, and 28 998. This comparison leads to the preliminary assessment of sentiment polarity. Should this step fail to meet the criteria, further sentiment analysis is carried out, as explicitly detailed in Vertical Detection. The implementation logic employed for the variable: **eval_outputs.logits** can be consulted in the code snippet for Vertical Detection, about the way conducting sentiment analysis based on the internal likelihood sorting outcome of the starting position of the aspect term.

Above is the breakdown (detailed analysis) of the *Affective Words Mapping* strategy about its detailed code implementation and concrete running-time logic. We hope that this analysis will assist readers in gaining a better understanding of the specific decoding mechanism of WordTransLM.

Appendix C. Case studies: Running-time WordTransLM sentiment analysis

In order to facilitate a better understanding of the intricacies of WordTransABSA for successive researchers, we select two typical sentiment prediction cases (the first two instances) from the development set of SemEval 2016 to conduct an in-depth analysis of the sentiment recognition process in WordTransABSA.

The runtime snapshots corresponding to the sentiment prediction procedure are presented in Fig. B.11. For simplicity, we opt for WordTransABSA$_{avg}$ for analysis, which is akin to several other WordTransABSA variants, making it a representative choice. In the WordTransABSA$_{avg}$, during the process of applying sentiment substitution tokens to the WordTransLM strategy, three sentiment representation vectors – namely, I-POS, I-NEU, and I-NEG – are introduced. In contrast, in other model variants like the WordTransABSA*clust*, these substitution tokens are derived from existing words within the PLM vocabulary.

For WordTransABSA$_{avg}$, the substitution tokens obtained through the *Average Pooling* strategy, I-POS, I-NEU, and I-NE, were manually added to the PLM vocabulary since they do not exist in the native PLM word vocabulary, as shown in sub-figure **2 B** of Fig. B.11. Specifically, We appended these substitution tokens to the end of the PLM vocabulary, resulting in their respective indexes being set to 28 996, 28 997, and 28 998.

Firstly, as illustrated in sub-figure **1 A** of Fig. B.11, these two inference cases are as follows: "The duck confit is always amazing, and the foie gras terrine with figs was incredibly delicious." and "I/we will never go back to this place again." In the first case, positive sentiment is expressed towards the aspect terms "duck confit" and "foie gras," while in second case, negative sentiment is expressed towards the aspect term, "this place.".

Subsequently, as depicted in sub-figure **3 C**, utilizing the training paradigm of Masked Language Modeling, Word Transferred LM directly generates the corresponding masked prediction sequence results,

```
for sent_idx, (pred_tag, real_tag) in enumerate(zip(pred_tags, real_tags)):
    for token_idx, (p_tag, r_tag) in enumerate(zip(pred_tag, real_tag)):
        if r_tag != 0:
            all_real_tags.append(r_tag)  # real senti-tag.
            if p_tag in tokenid2label:
                final_tag = label2id[tokenid2label[p_tag]]
                all_pred_tags.append(final_tag)
                situation_record['senti_first_appear'] += 1
            else:
                if args.answer_strategy == 'inter_logits_core_token':
                    # Find the emotion labels in the current location's emotion label set
                    # that have a high probability of matching.
                    all_probs = eval_outputs.logits[sent_idx, token_idx]
                    pred_senti_probs = torch.index_select(all_probs, 0, index_tensor).detach().cpu().tolist()
                    # logger.info(pred_senti_probs)
                    max_senti_prob = -999
                    final_tag = 0
                    for senti_prob, sub_tokenid2id in zip(pred_senti_probs, tokenid2id):
                        if senti_prob > max_senti_prob:
                            max_senti_prob = senti_prob
                            final_tag = sub_tokenid2id[1]
                    if not final_tag > 0:
                        logger.error(f'final_tag = 0: pred_senti_probs: {pred_senti_probs} tokenid2id: {tokenid2id}')
                    all_pred_tags.append(final_tag)
```

Horizontal Detection

Vertical Detection

Fig. B.9. The code logic of affective words mapping for decoding sentiment polarity, specifically the pipeline combination of horizontal detection and vertical detection.

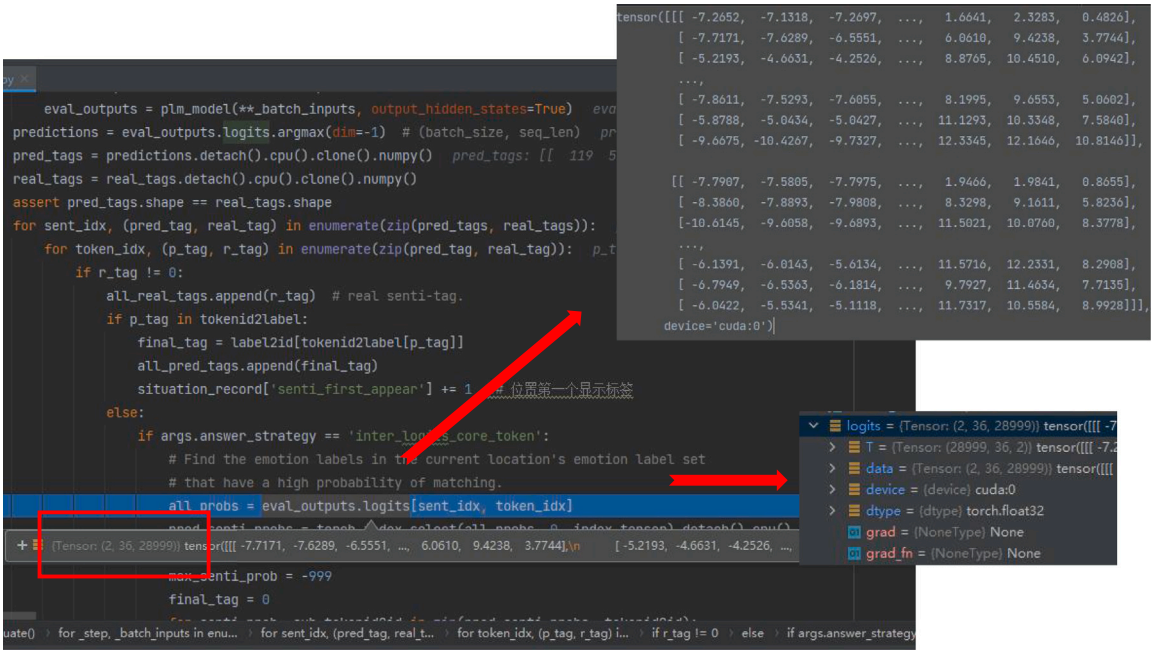


Fig. B.10. Runtime visualization of logits tensors output by Word Transferred Language Model directly (corresponding to the program variables: **eval_outputs.logits**).

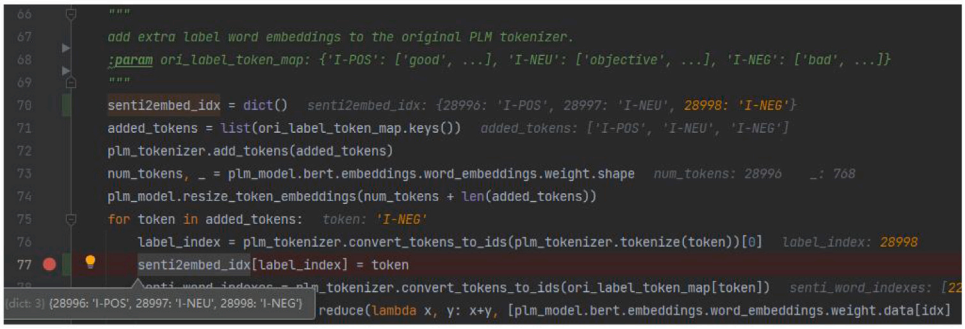
denoted as variable: **eval_outputs.logits** in this code snippet. For an in-depth discussion of **eval_outputs.logits**, Appendix B give a detailed description, which represents a probability distribution for each position in the transferred linguistic sequence. This distribution indicates the likelihood of each word belonging to the PLM word vocabulary. It is worth noting that these logit values vary across the real number spectrum.

Furthermore, as depicted in sub-figure **4 D**, we firstly employ Horizontal Detection to analyze these logits. We utilize the function '.argmax(dim=-1)' to retrieve the predicted word with the highest probability at the position corresponding to the aspect term. Subsequently, we determine the sentiment polarity by assessing whether the predicted word belongs to the sentiment-related pivot tokens.

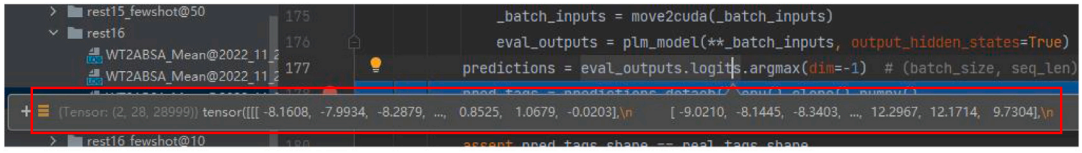
During the sentiment ploarity inferences, Horizontal Detection can effectively handle most context scenarios, directly identifying sentiment substitution tokens, I-POS, I-NEU, and I-NEG. As illustrated in sub-figure **5 E**, Take the first case as an example, WordTransLM predicts the substitution token indices for positions 2, 3, and 11 (corresponding to the entity words "duck confit" and "terrine") as 28 996, indicating the substitution token I-POS. Consequently, WordTransABSA can determine the positive sentiment polarity for these two entities. Similarly, for the second case, WordTransLM correctly predicts that the



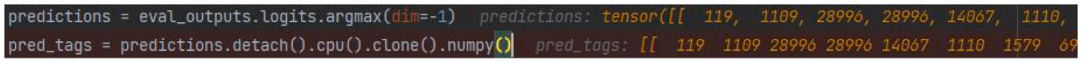
1 A. The inference case studies (two cases) which is the first two validation instances from SemEval 2016 development set.



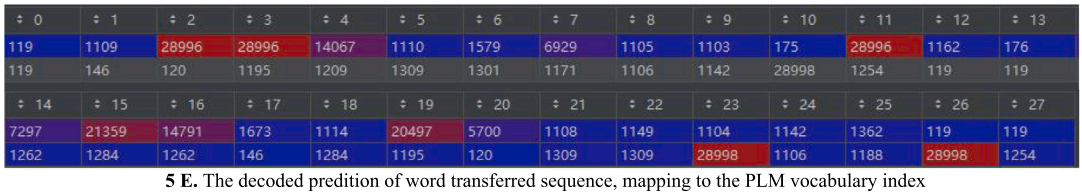
2 B. The three sentiment substituted embeddings (idx:28996,28997,28998) that represent POS / NEU / NEG.



3 C. The **logits** value that generated from Word Transferred LM, **size**: (batch size, sequence length, vocab size).



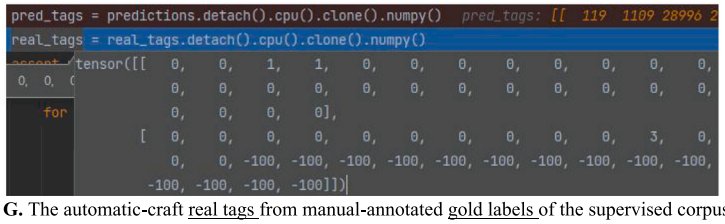
4 D. The SoftMax operation filtering the maximum probability word, corresponds to *Horizontal Detection*.



5 E. The decoded predition of word transferred sequence, mapping to the PLM vocabulary index , such as 28996 (for first case) and 28998 (for second case).



6 F. The sentiment polarity tag which maps to the pre-set sentiment index.



7 G. The automatic-craft real tags from manual-annotated gold labels of the supervised corpus.

Fig. B.11. Case analysis about the specific execution process of sentiment identification in Word Transferred Language Model of WordTransABSA.

position corresponding to the entity "place" belongs to the sentiment substitution token I-NEG, successfully identifying "place" as having a negative sentiment polarity. Due to the influence of PLM's padding and sequence tokenization pre-processing, some target entities may not directly correspond between their positions in the original sentence and the decoding sequence, in which the specific positional mapping has been implemented in our code.

Sub-figure **6 F** illustrates the mapping of pre-set sentiment indices to sentiment polarity tags, facilitating the alignment of aspect terms with their respective sentiment labels for subsequent accuracy performance assessment with the actual labels. Sub-figure **7 G** corresponds to the true labels for these two cases. Notably, the true labels do not explicitly denote a positive sentiment for entity word "terrine", yet WordTransABSA adeptly recognizes it.

The above offer a comprehensive analysis towards a concrete sentiment inference of WordTransABSA. Through this discussion, readers can be benefited by gaining a profound insight into the operational mechanism of WordTransABSA from it.

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